

① Friction Electricity :

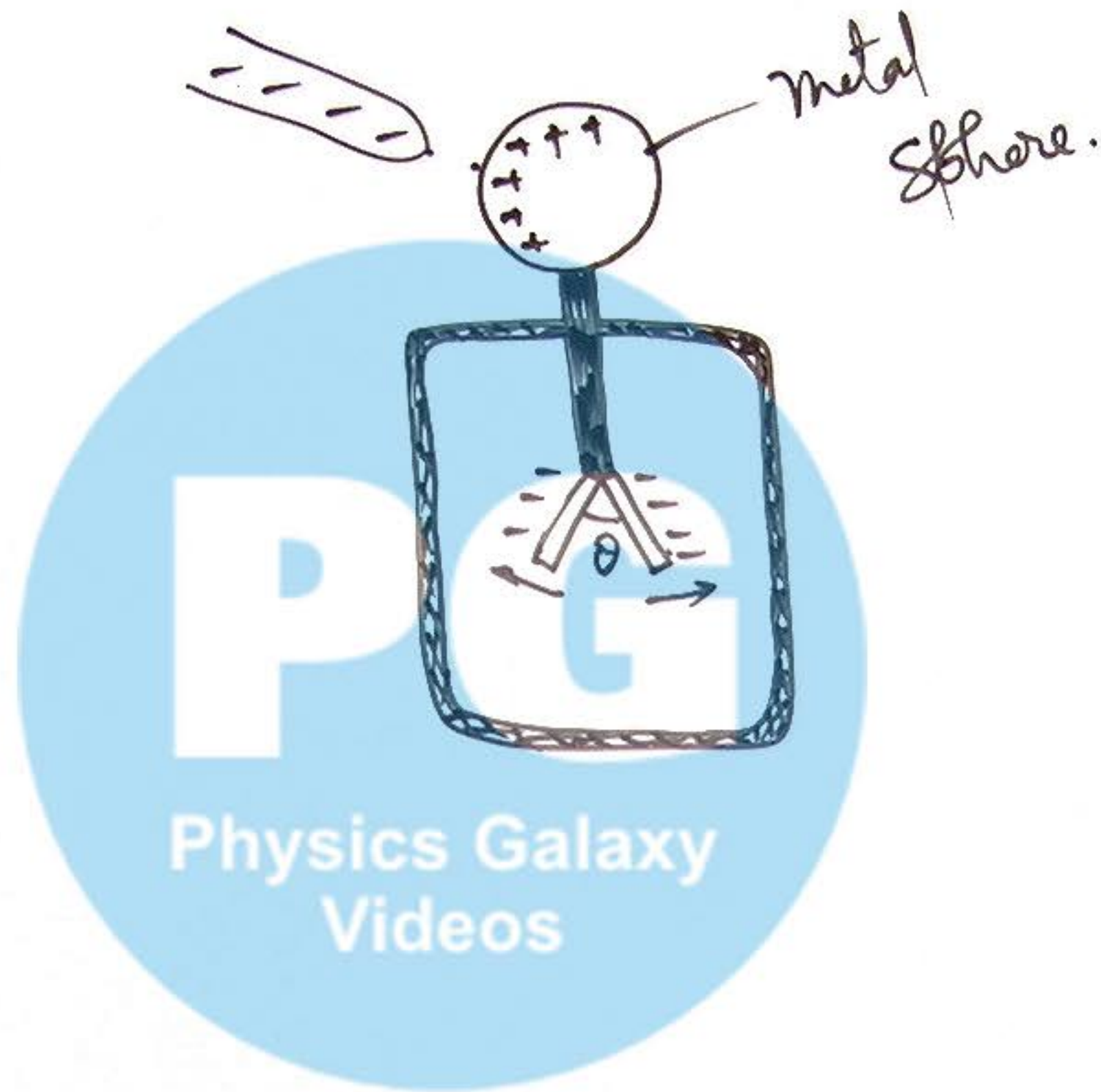
↓
It works only when
at least one body
is Insulator.



PG

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② Gold Leaf Electroscope :

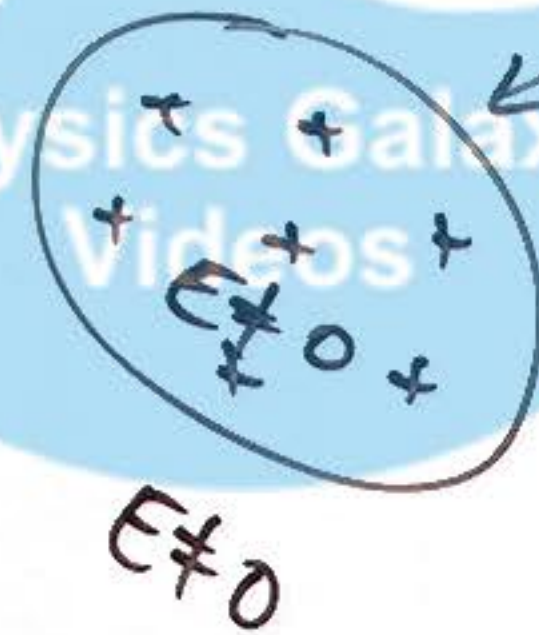
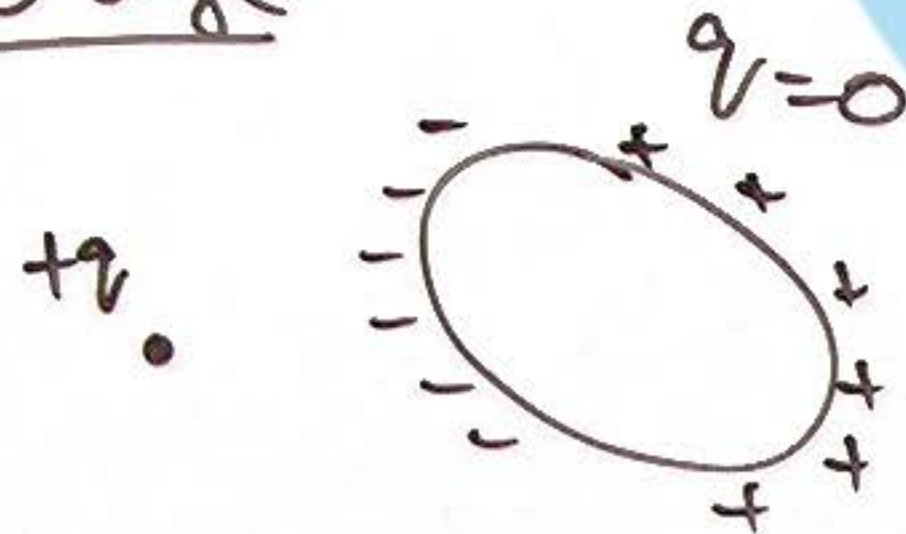


③ Properties of Electric Charge:→ quantization

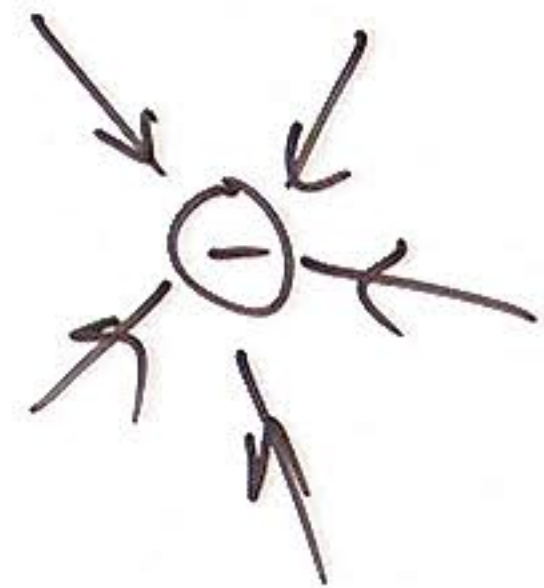
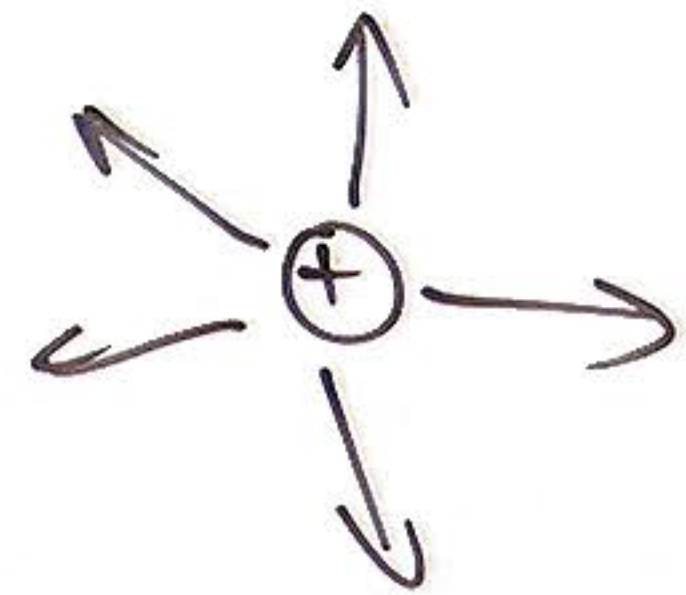
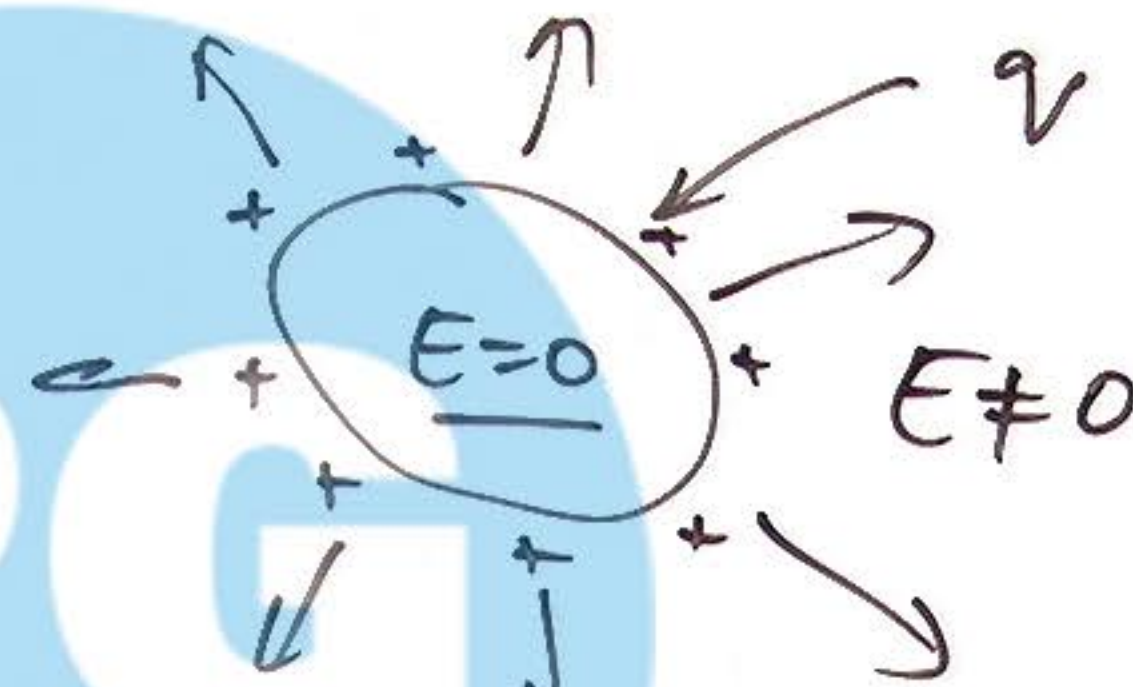
$$q = Ne \quad (NEI)$$

$1.6 \times 10^{-19} \text{ C}$

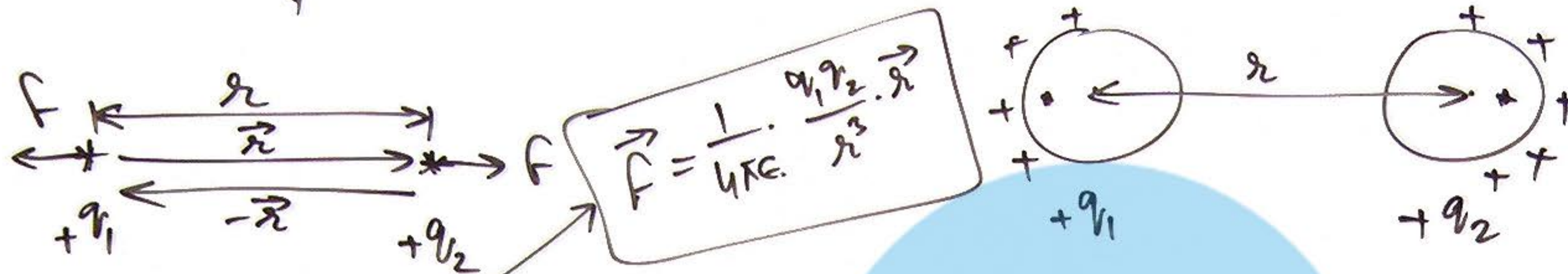
charging
 $\Rightarrow$
 addition / removal of e^-

Conducting Bodies (Metal)Non-Conducting (Insulator)Induction of charge

uniformly / non-uniformly



④ Coulomb's Law & Superposition:



$$F = K \frac{q_1 q_2}{r^2}$$

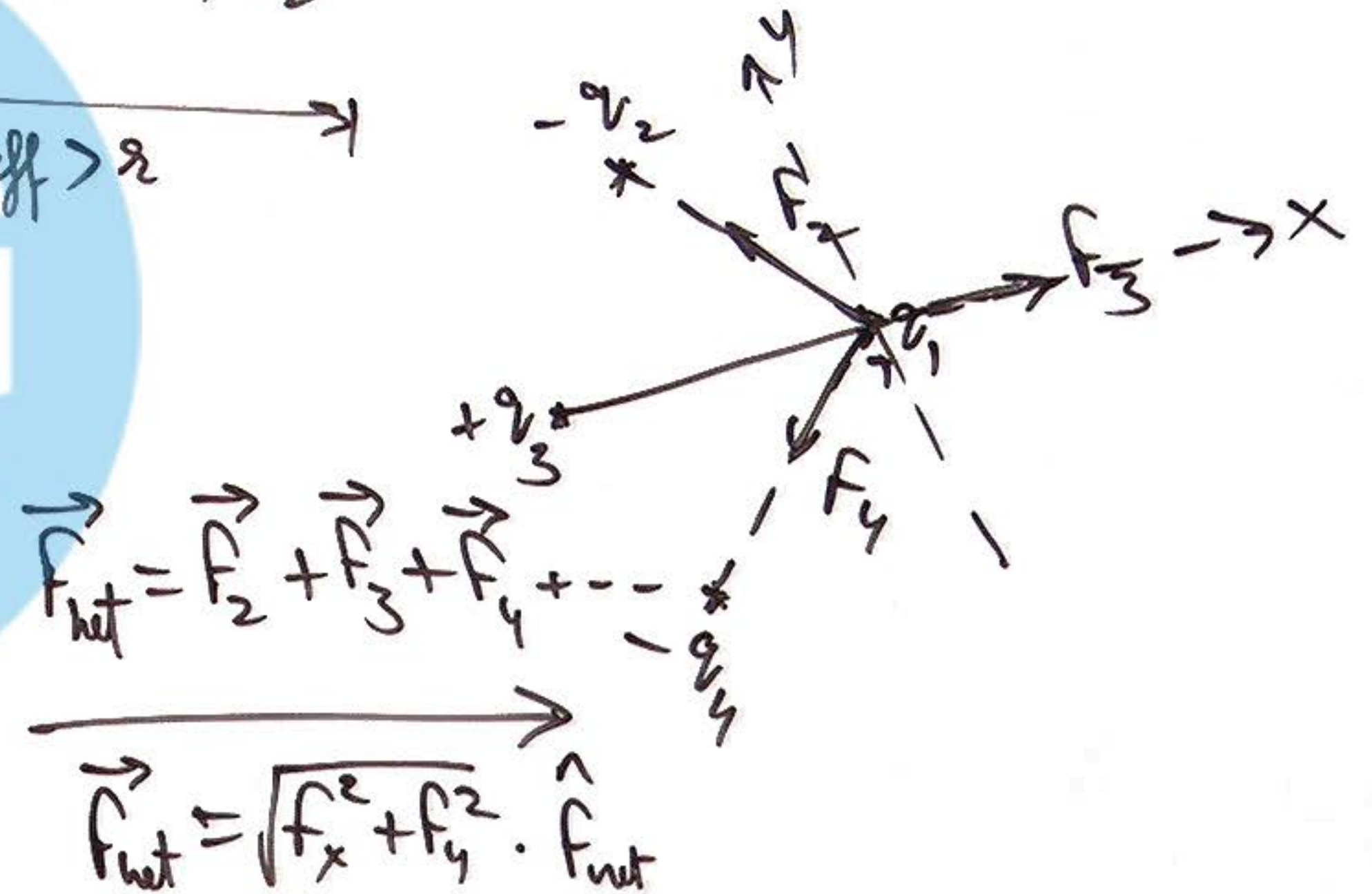
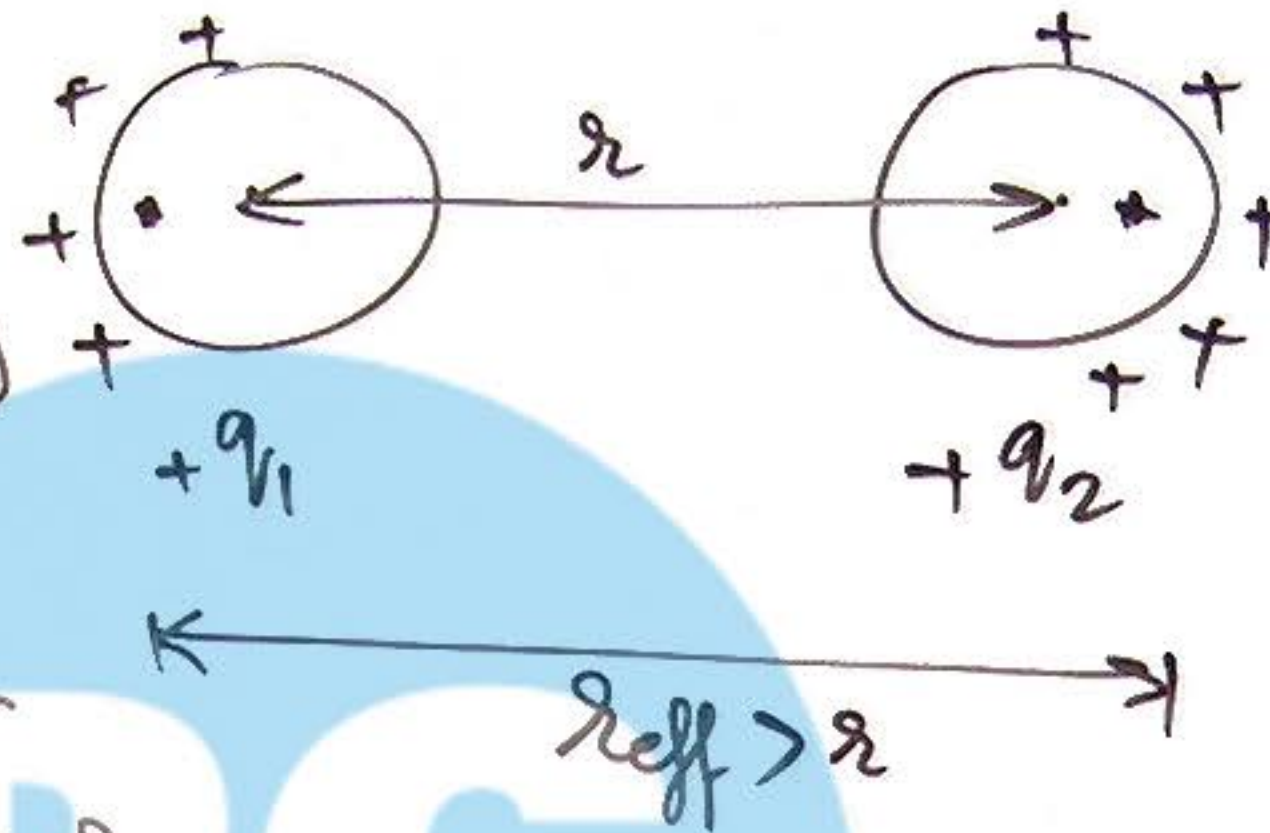
$$\Rightarrow F = \frac{1}{4\pi\epsilon} \cdot \frac{q_1 q_2}{r^2}$$

$\epsilon = \epsilon_0 \cdot \epsilon_r$
 ϵ_0 → perm of free space.
 ϵ_r → dielectric const / relative perm.

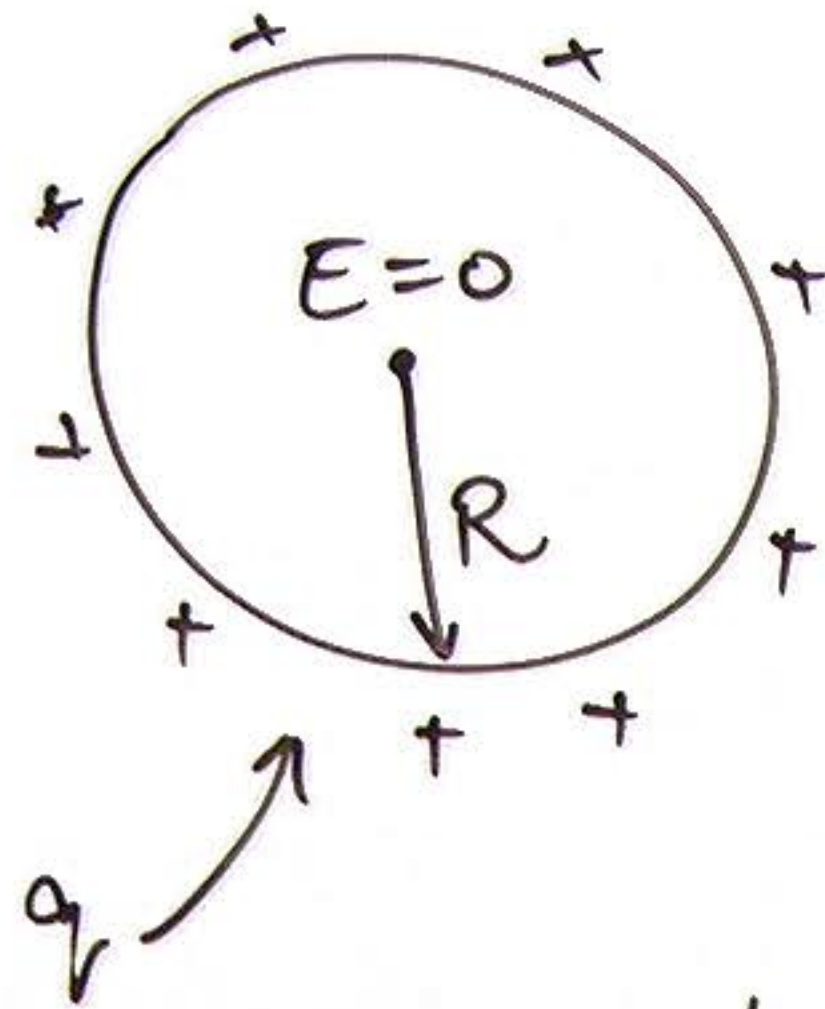
Limitation

→ L1 → only for point charges

→ L2 → both charges must be at rest



⑤ Charge Distribution on an Isolated Conductor:



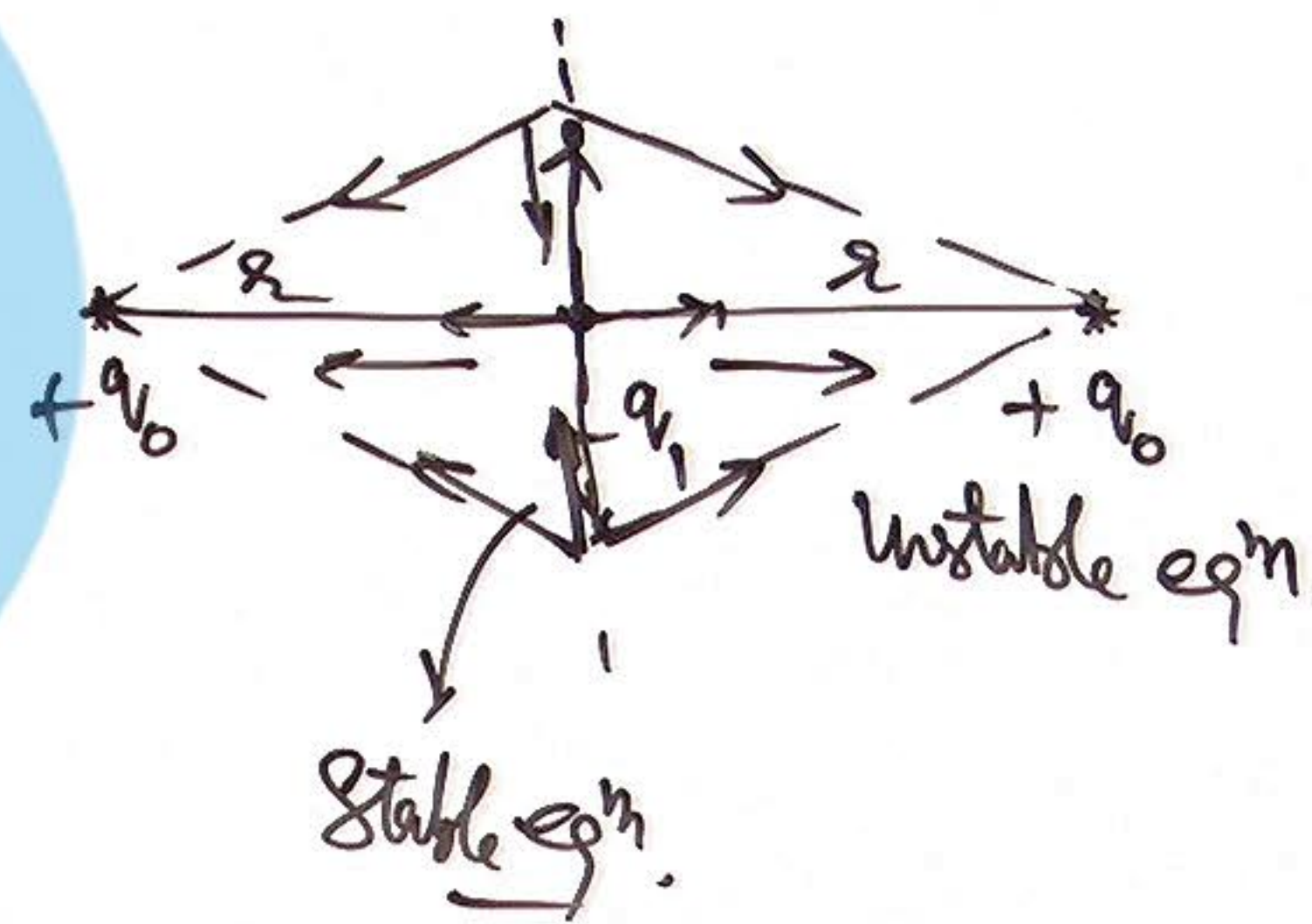
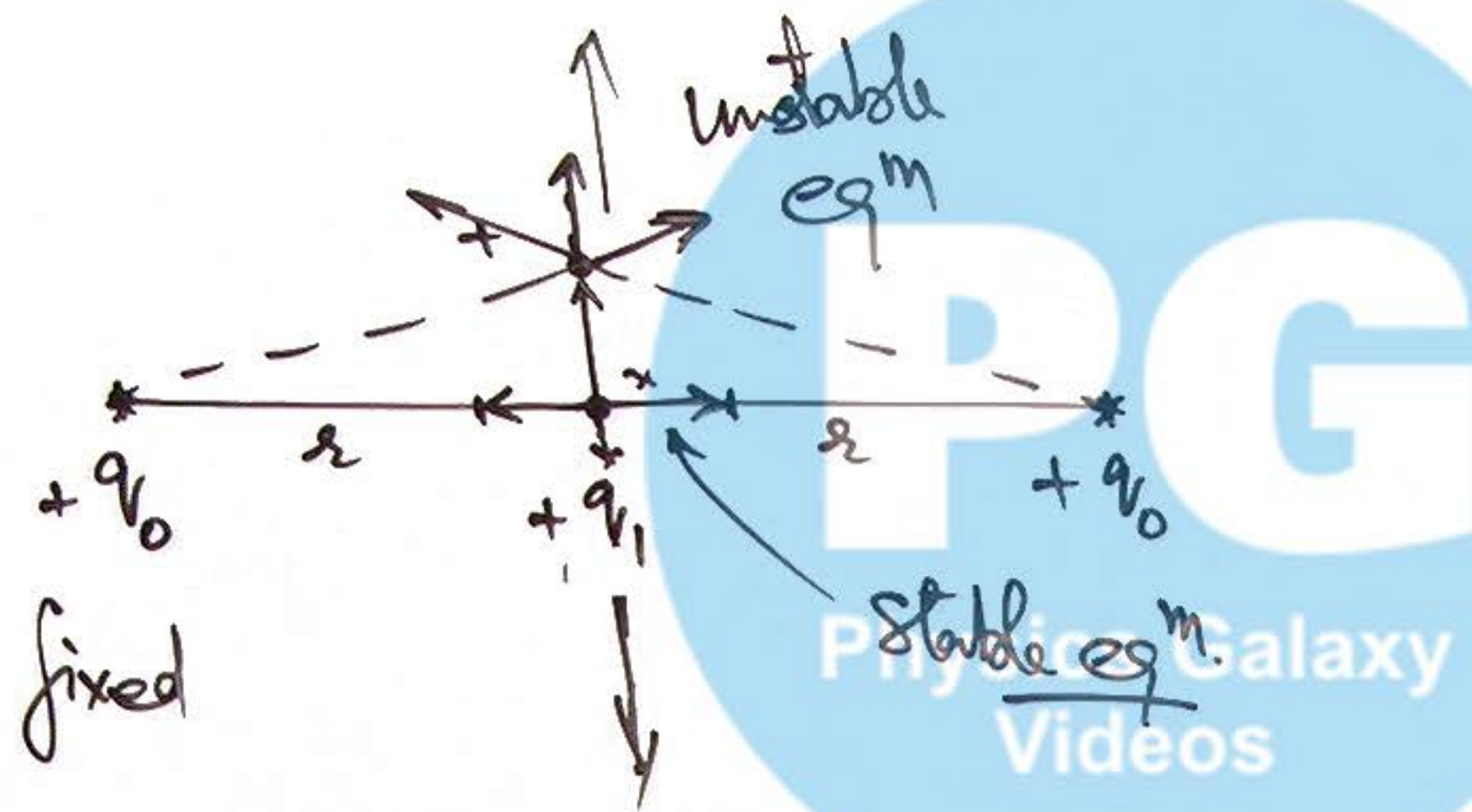
$E=0$ always inside
an isolated conductor

Surface charge density

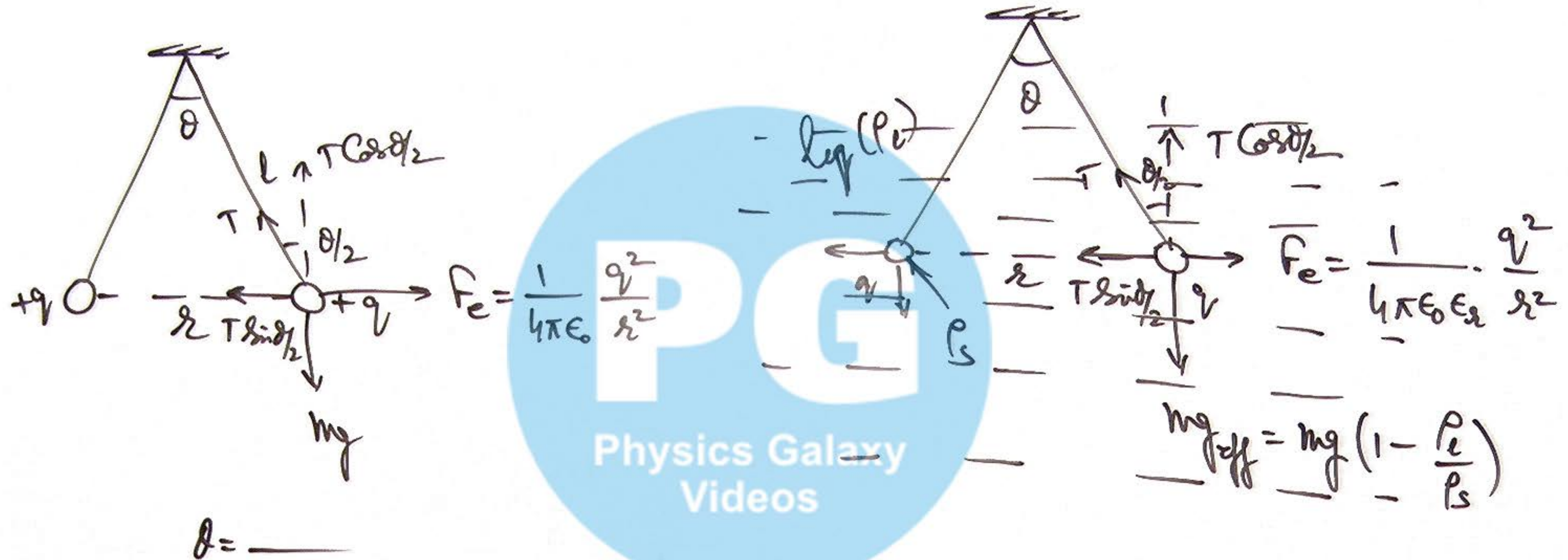
$$\sigma \propto \frac{1}{r}$$

for isolated conductor

⑥ Equilibrium of a charge between
Two fixed charges:



⑦ Equilibrium of two suspended charges:

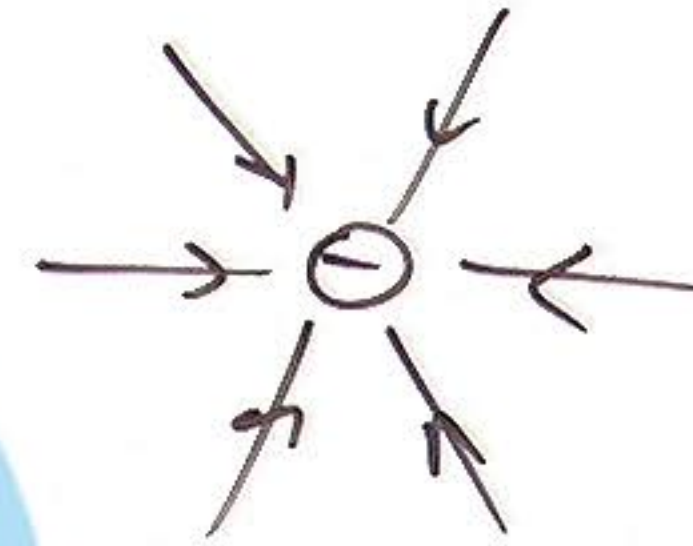


⑧ Electric Field:

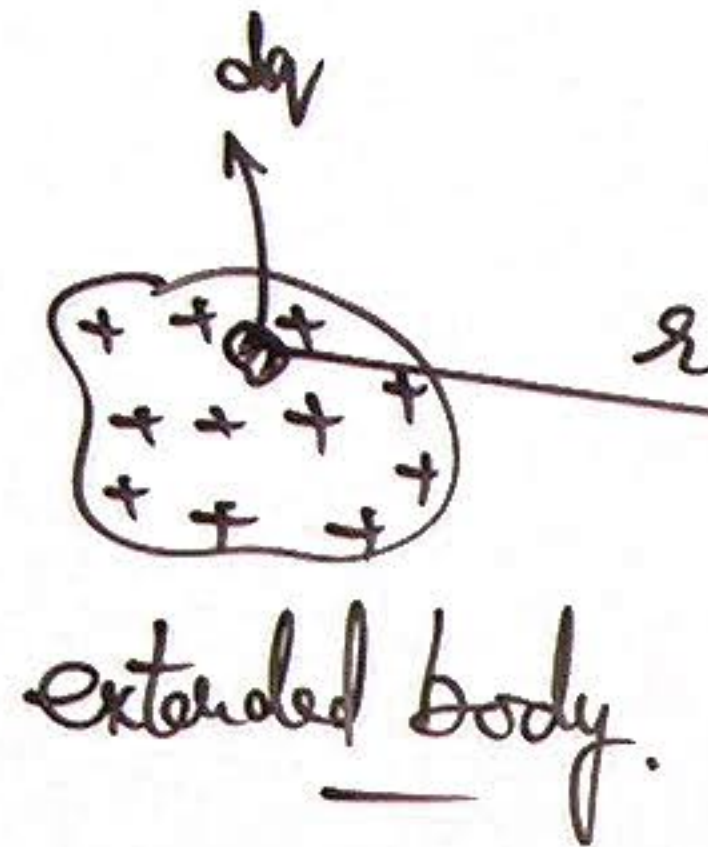
force per unit +ve charge.

it can exert force on static charge.

every charge produces EF



$$\vec{E} = E \cdot \hat{n}$$



Ef due to a point charge

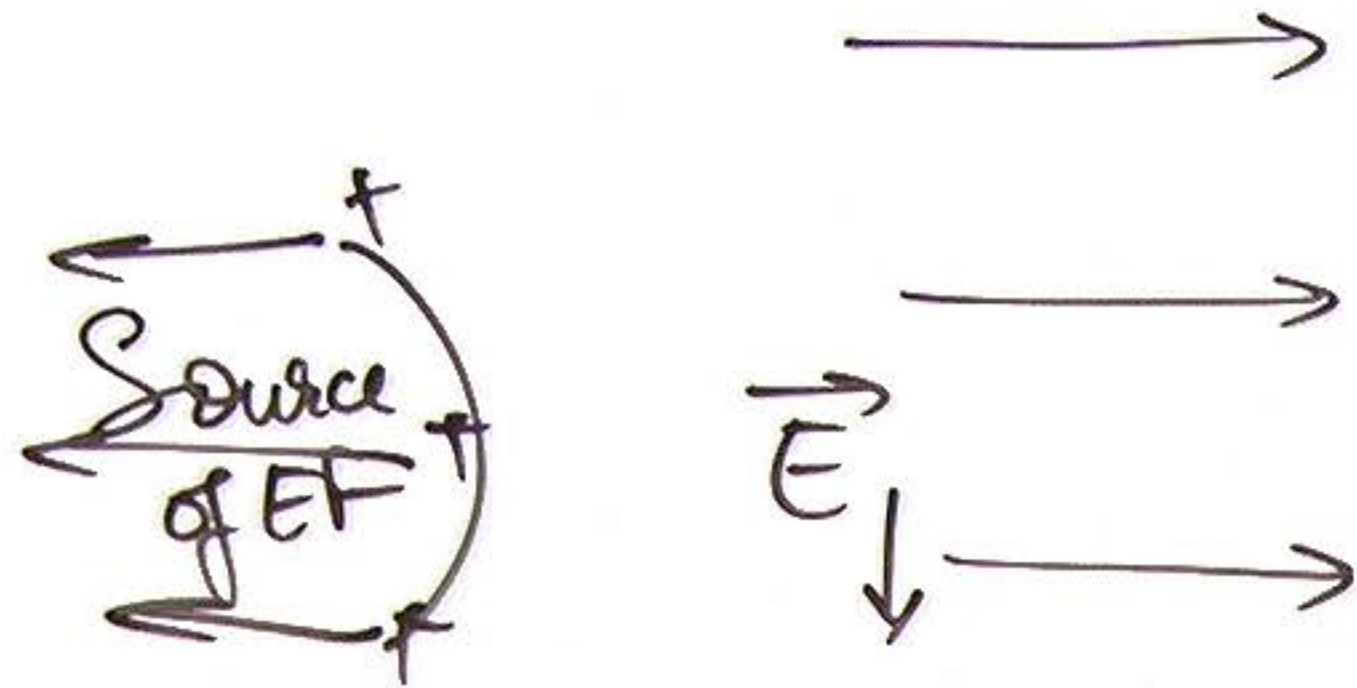
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$$dE = \frac{k dq}{r^2}$$

$$E = \frac{kq}{r^2}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$\vec{E} = \frac{kq}{r^2} \cdot \hat{r} = \frac{kq}{r^3} \vec{r}$$

⑨ Understanding a Test Charge:

$$\vec{E} \neq \frac{\vec{F}}{q}$$

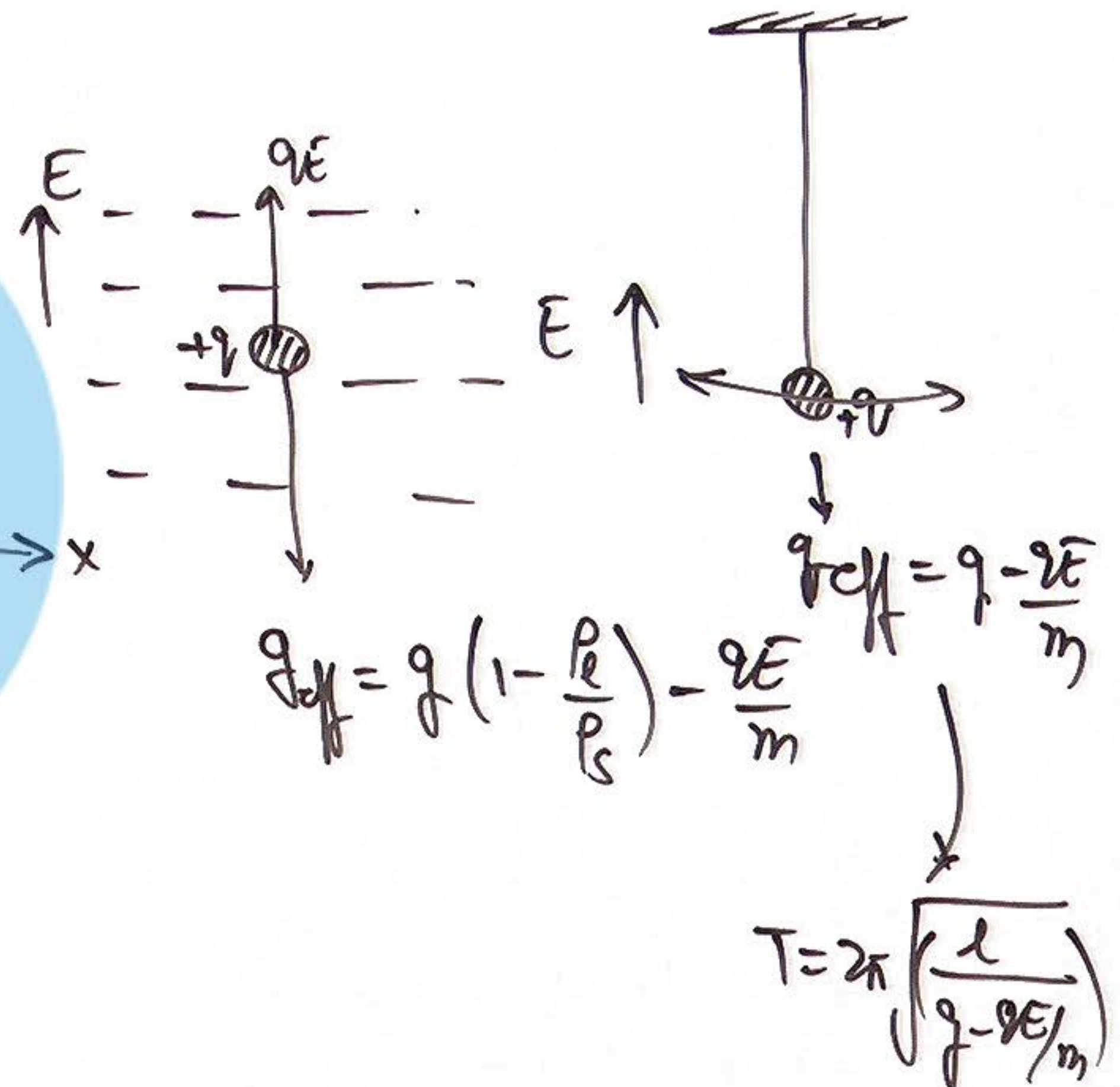
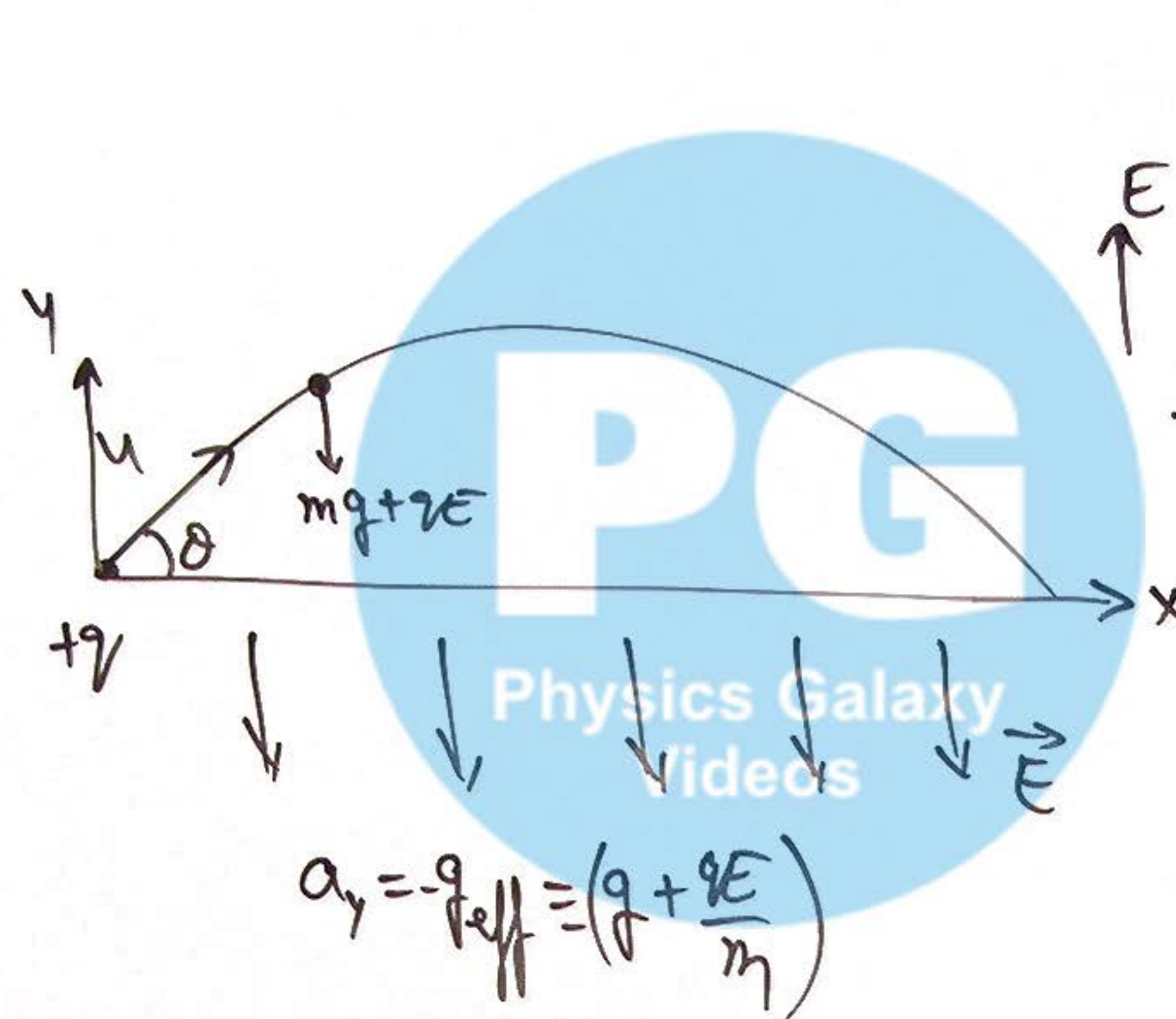
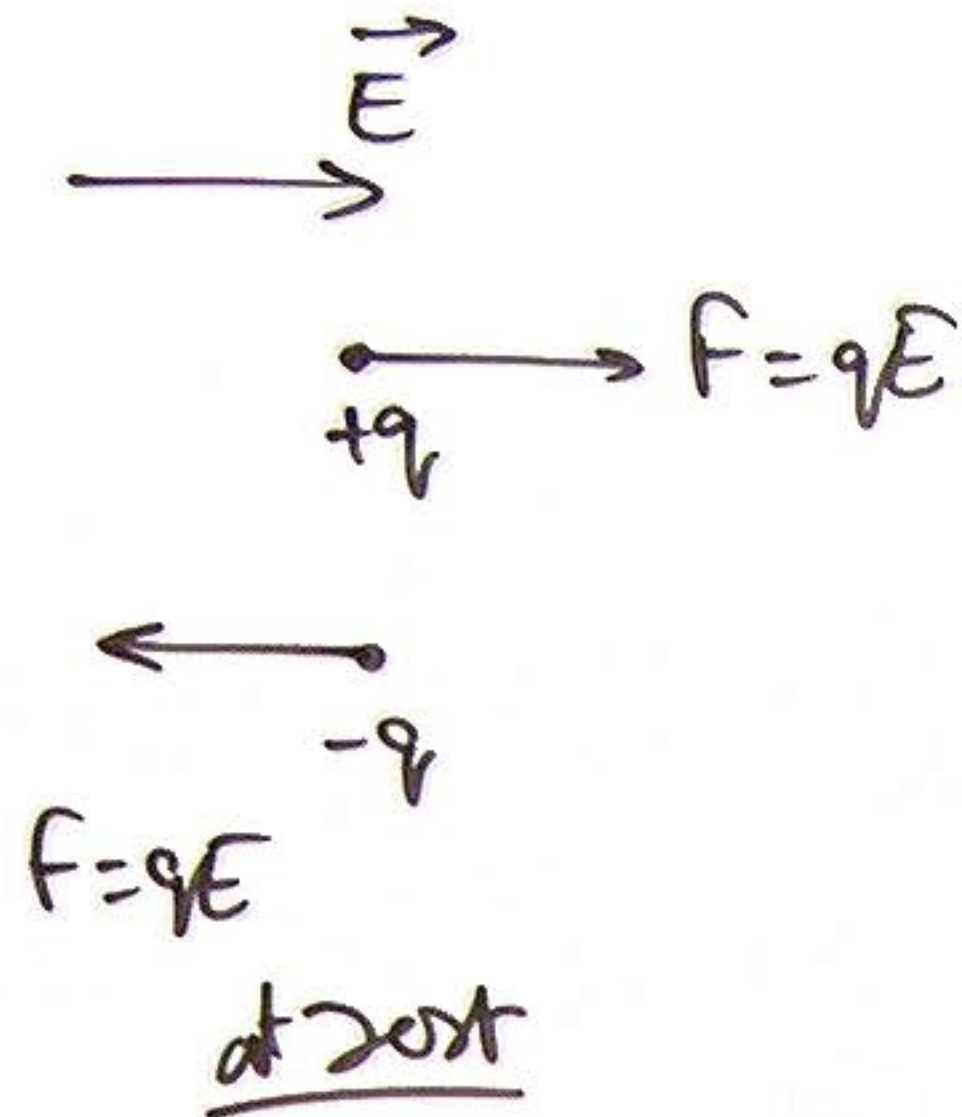
✓

$$\vec{E} = \lim_{q \rightarrow 0} \frac{\vec{F}}{q} = \frac{\vec{F}_0}{q_0}$$

→ test charge.

$+q$
↓
almost zero.

⑩ Effect of E on moving charges:



(11) 'Electric Flux' Calculation

& Gauss Theorem:

→ group of field lines

passing through a surface.

flux due to a uniform EF

$$\phi = \vec{E} \cdot \vec{S} = ES \cos \theta$$

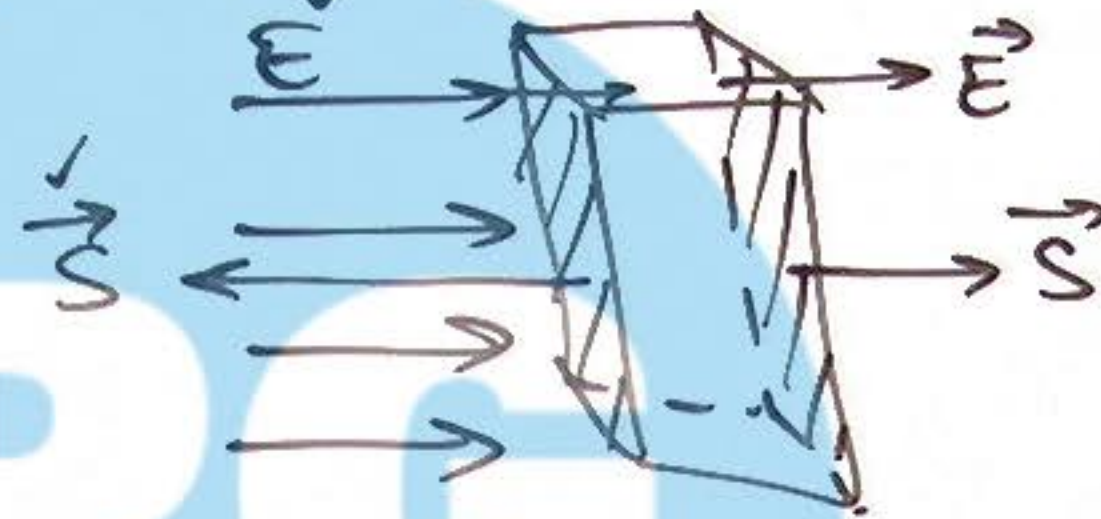
for non-uniform EF

$$\phi = \int \vec{E} \cdot d\vec{S} = \int E dS \cos \theta$$

passing through $\Rightarrow$ magnitude
 $\phi_{\text{net}} = |\phi_{\text{in}}| = |\phi_{\text{out}}|$

$$\phi_{\text{in}} = -ES$$

$$\phi_{\text{out}} = +ES$$



$$\phi = E \times 2RL$$

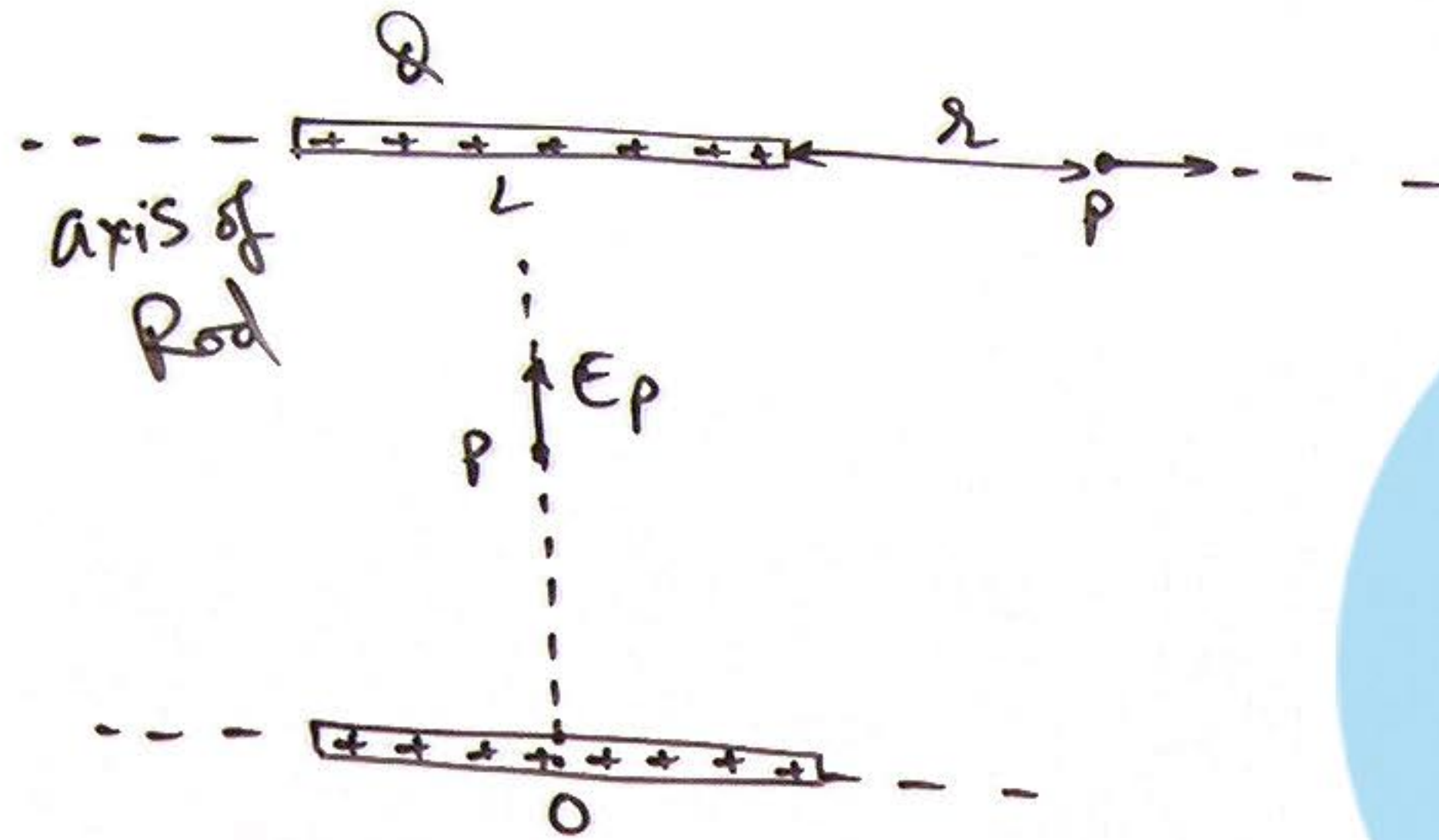
Gauss law.

for any closed surface.

$$\phi_{\text{net out}} = \frac{q_{\text{enc}}}{\epsilon_0}$$

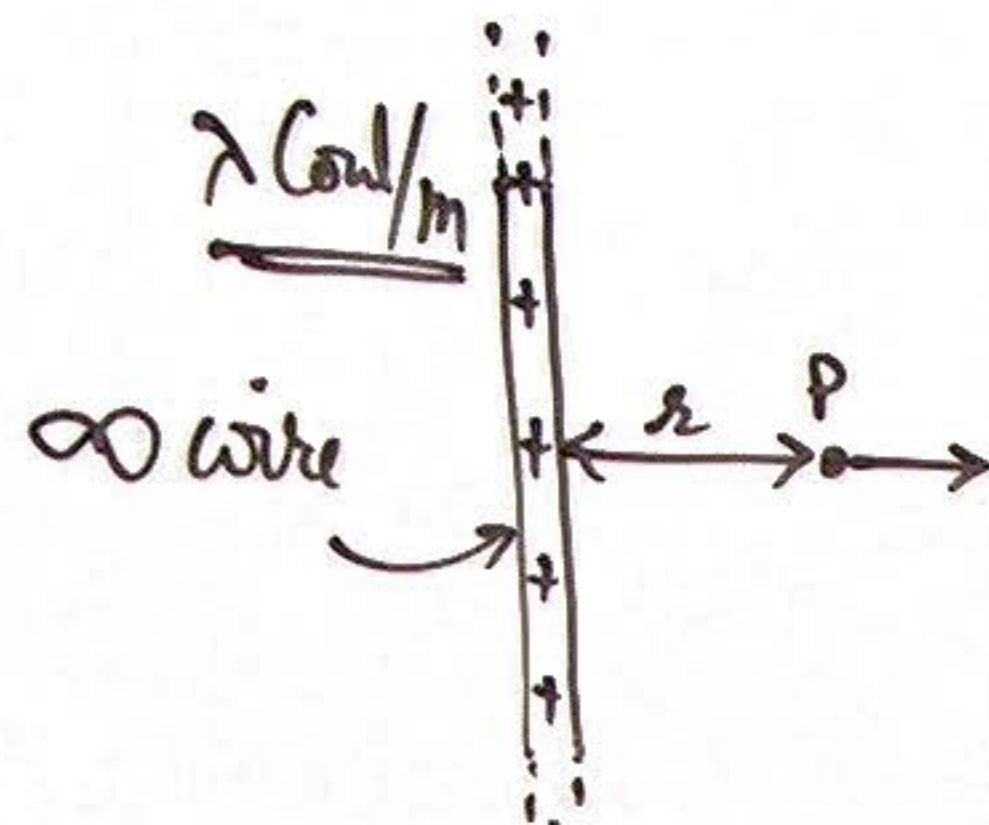
$$\oint \vec{E} \cdot d\vec{S} = \frac{q_{\text{enc}}}{\epsilon_0}$$

⑫ EF due to uniformly charged Rod:

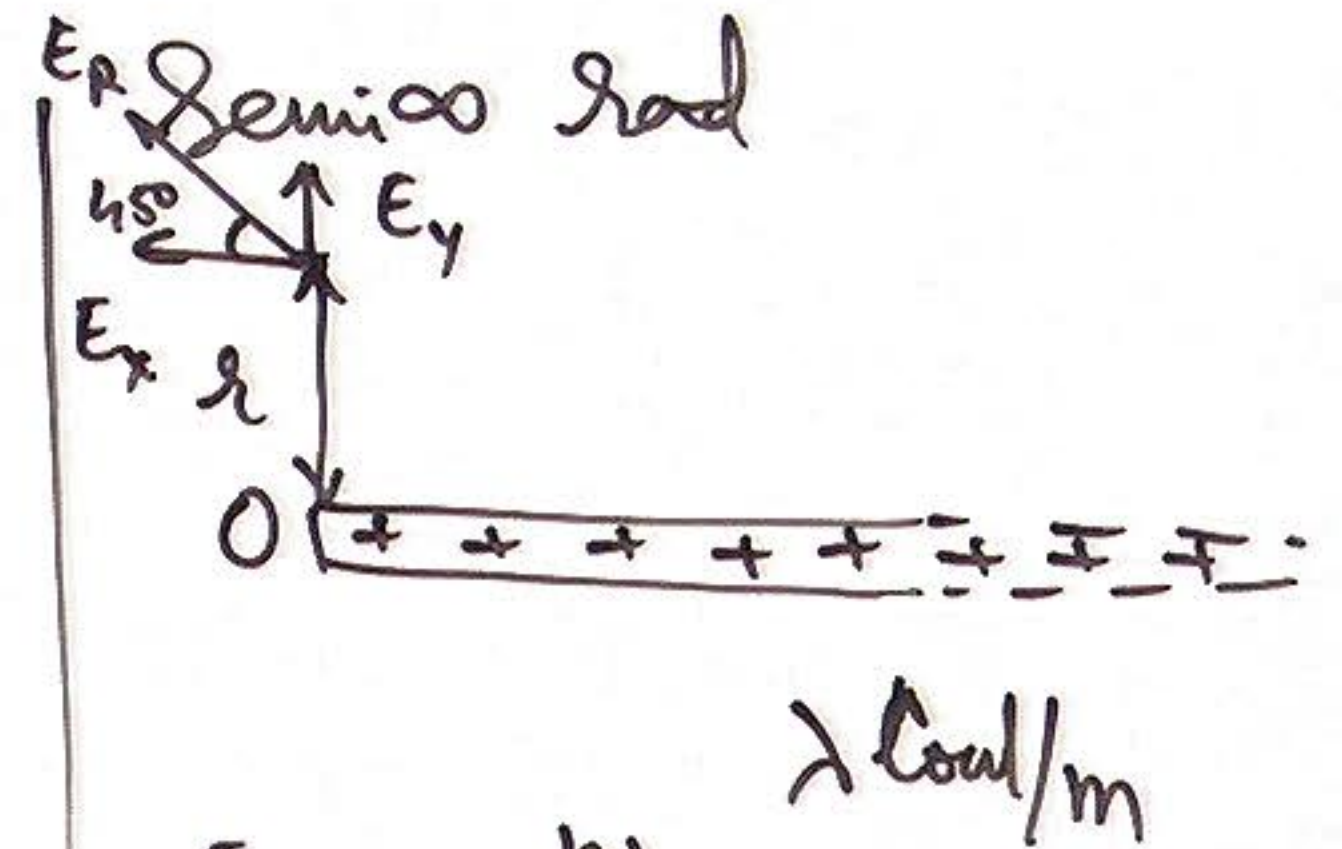


$$E_{P_{||}} = \frac{kQ}{L} \left[\frac{1}{r} - \frac{1}{r+L} \right]$$

$$E_{P_{\perp}} = \frac{2kQ}{r \sqrt{L^2 + 4r^2}}$$



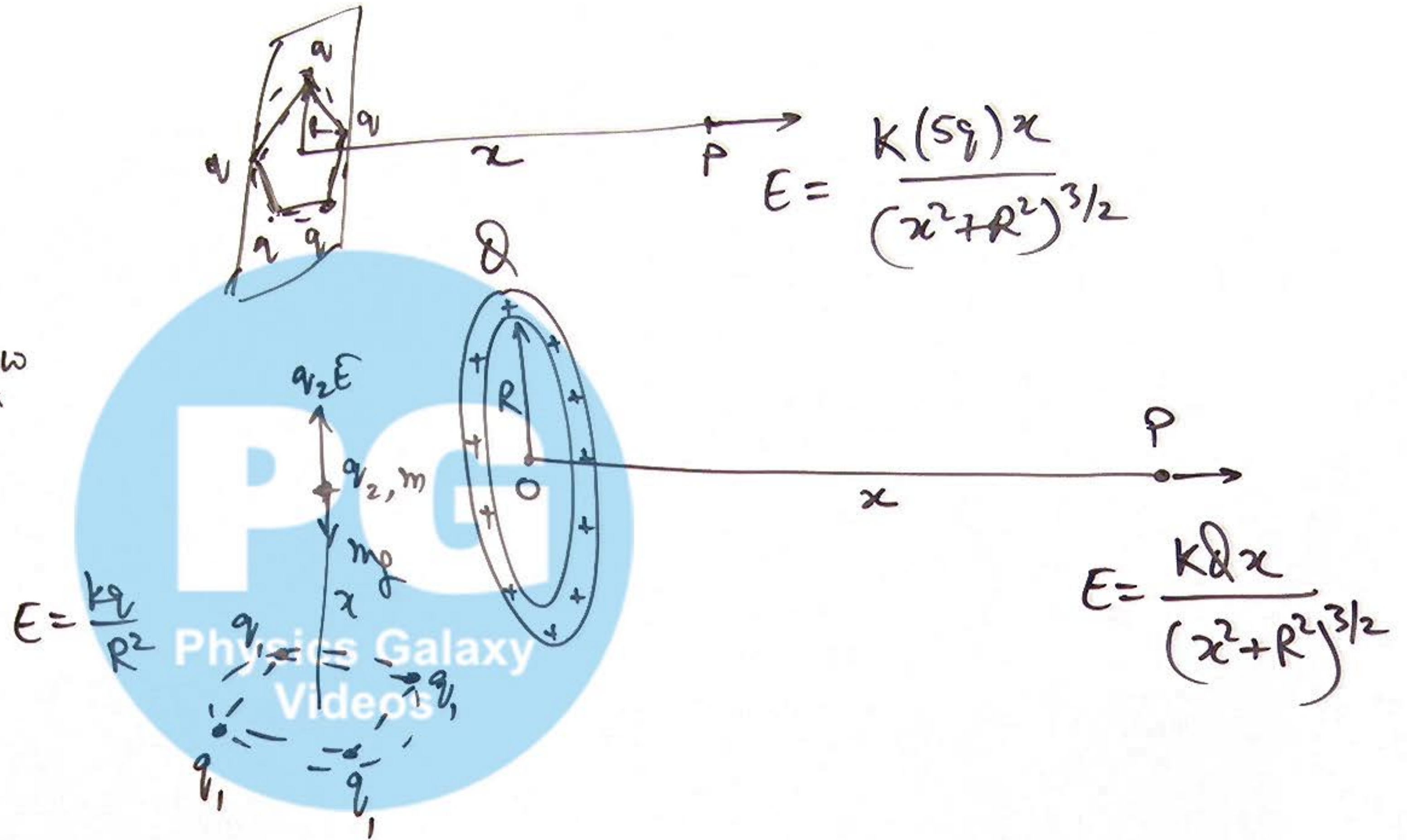
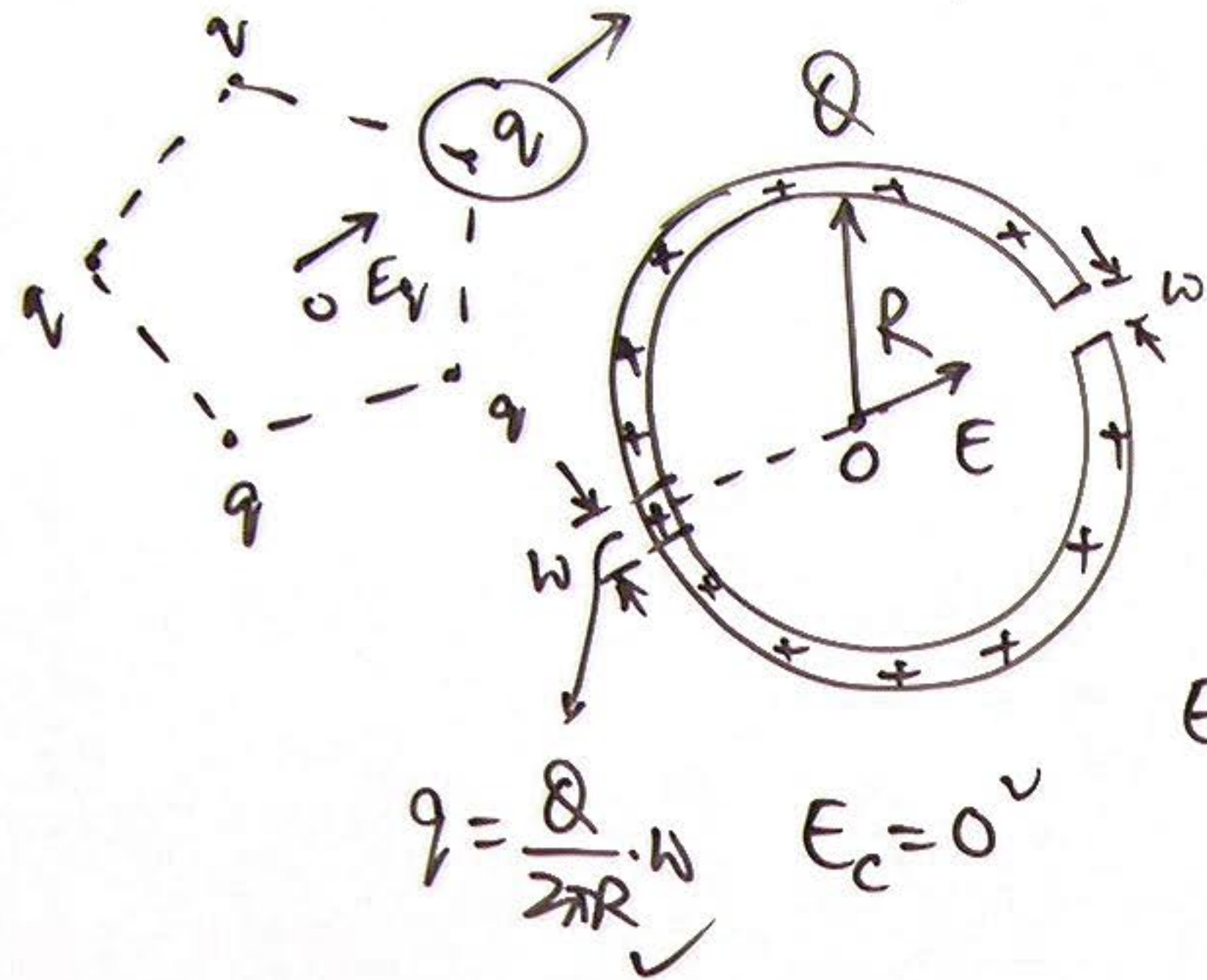
$$E_P = \frac{2k\lambda}{r}$$



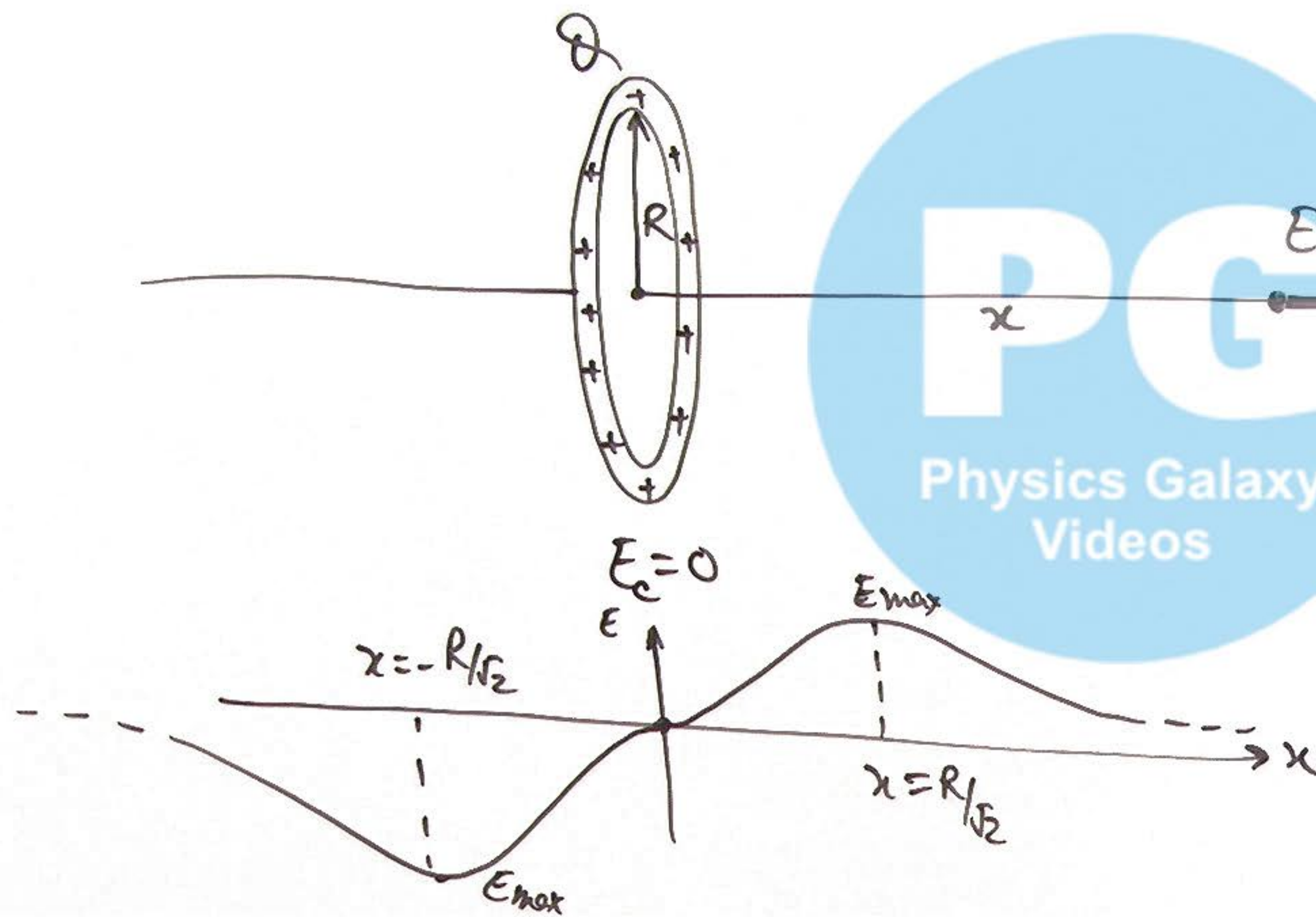
$$E_x = E_y = \frac{k\lambda}{r}$$

$$E_P = \sqrt{E_x^2 + E_y^2} = \sqrt{2} \frac{k\lambda}{r}$$

13) EF due to a uniformly charged Ring:



14 Variation of EF on Ring Axis :



$$E_x = \frac{KQx}{(x^2 + R^2)^{3/2}}$$

E is max when

$$\frac{dE}{dx} = 0$$

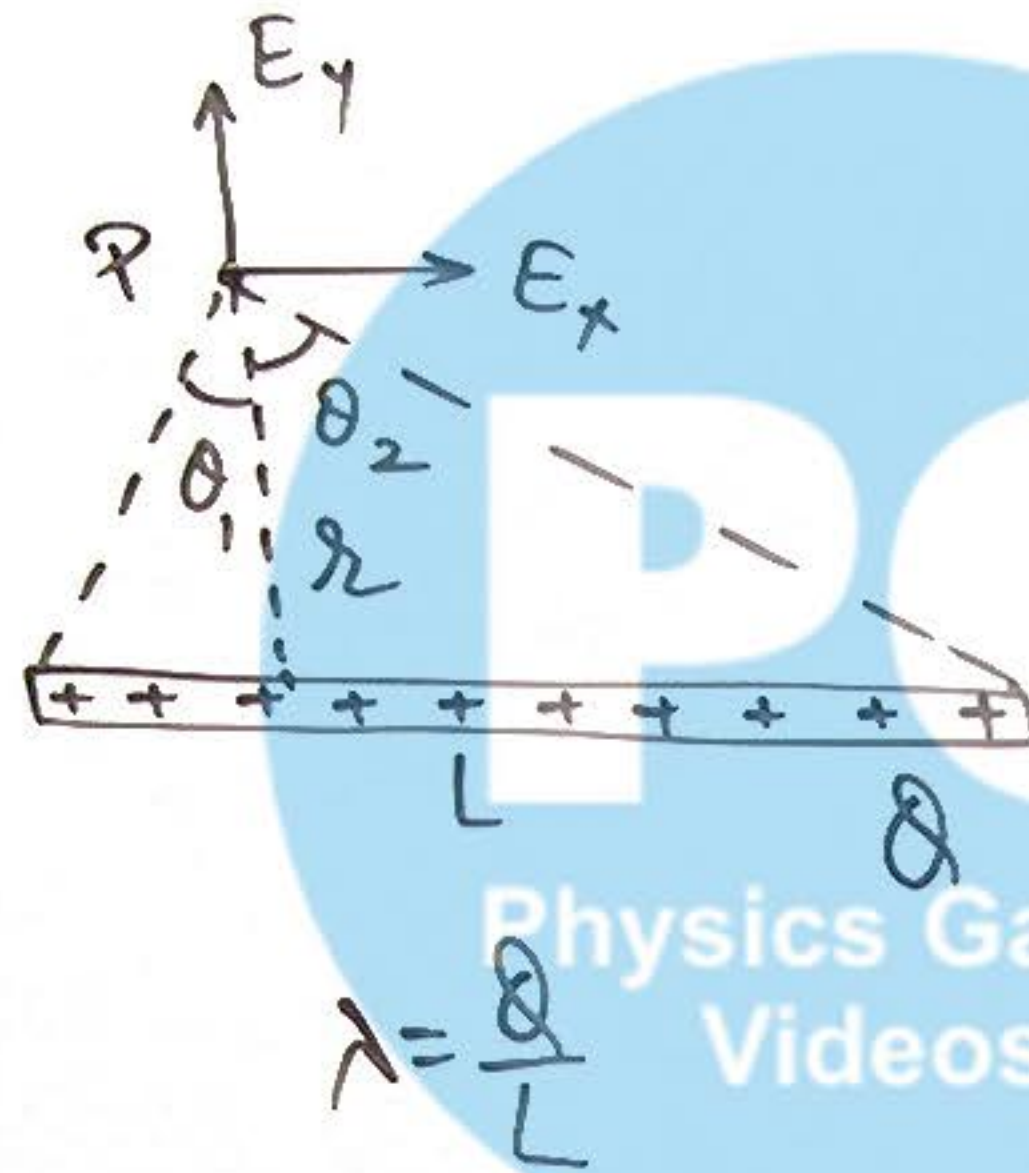
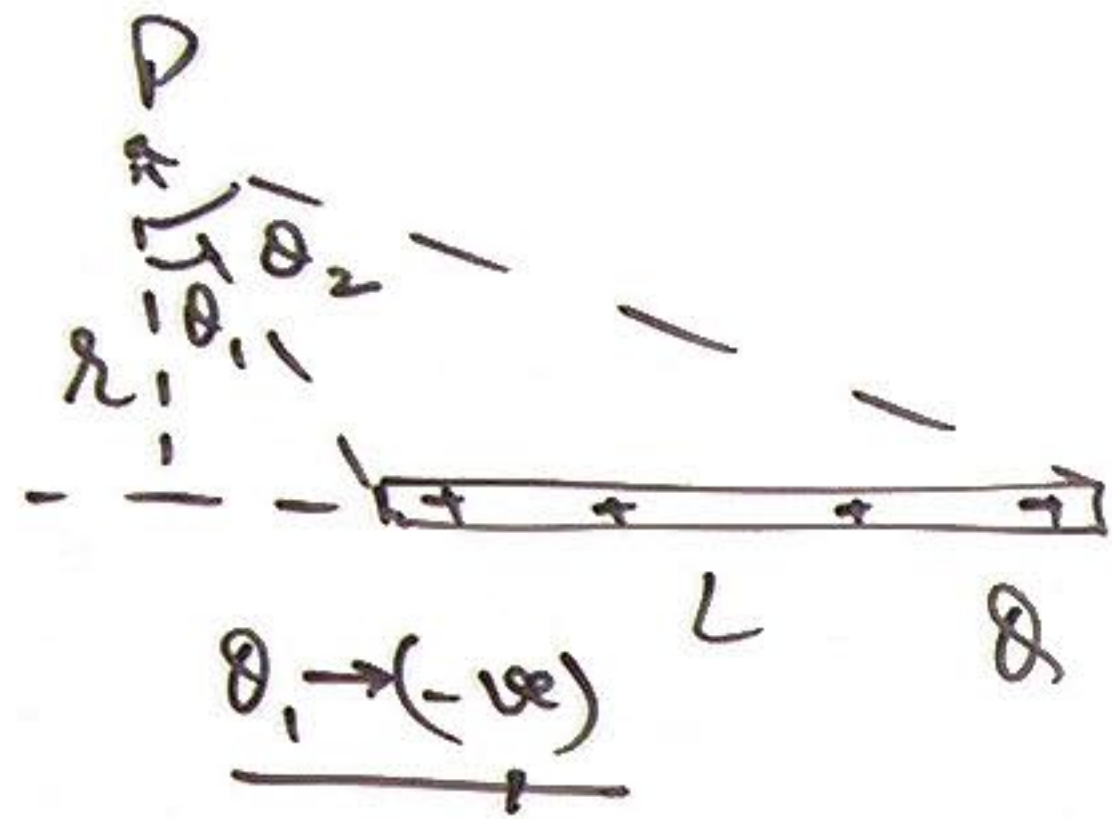
$$\Rightarrow x = \pm \frac{R}{\sqrt{2}}$$

$$E_{\infty} = 0$$

$$E_{max} = \frac{2KQ}{\sqrt{27} R^2}$$

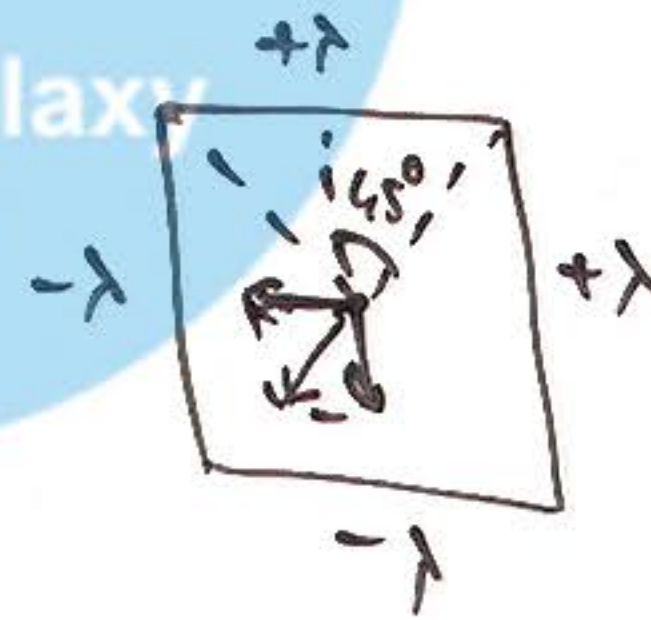
at $x = \pm R/\sqrt{2}$

(15) EF due to a finite length wire:

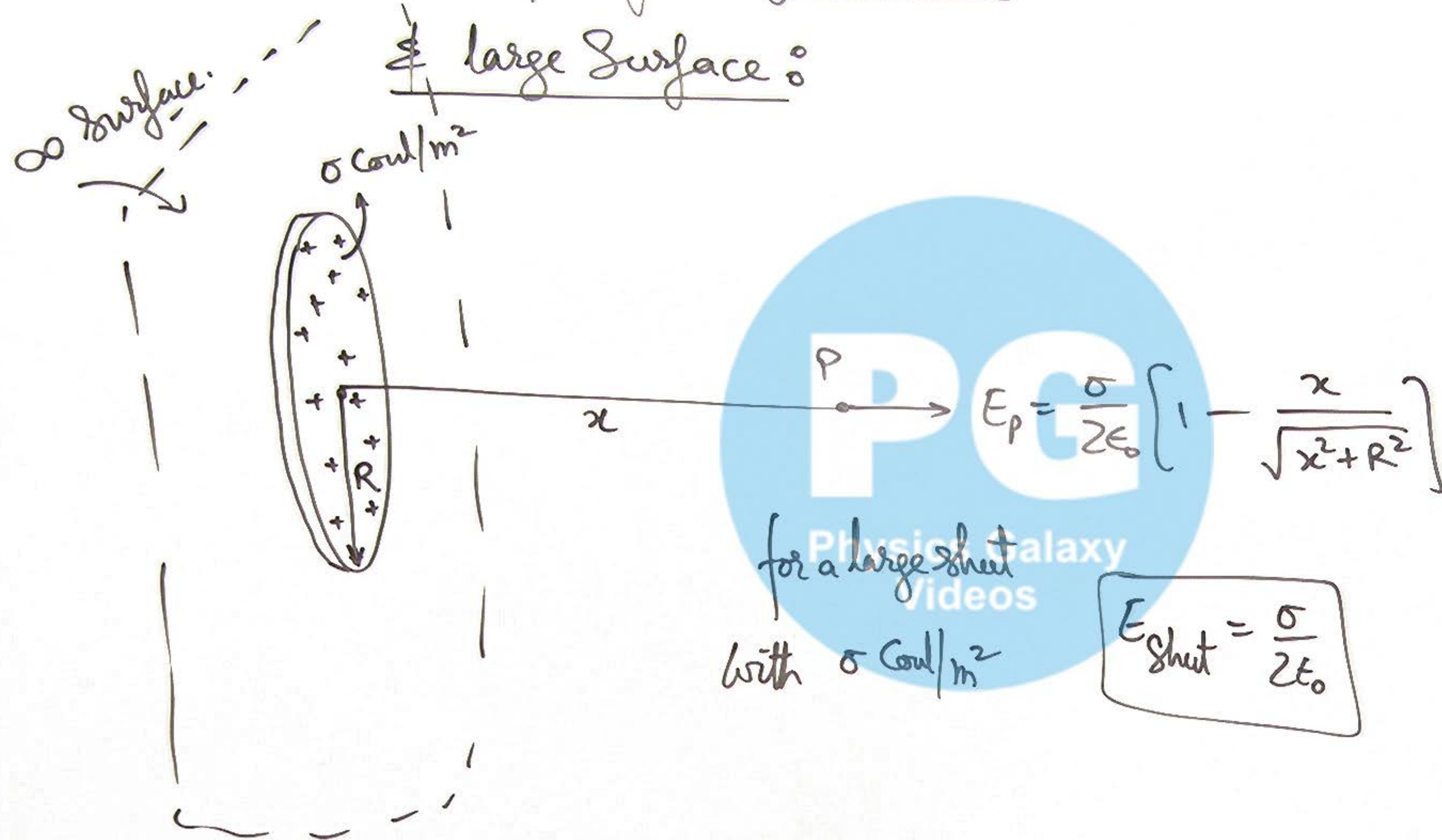


$$E_x = \frac{k\lambda}{r} [\cos\theta_2 - \cos\theta_1]$$

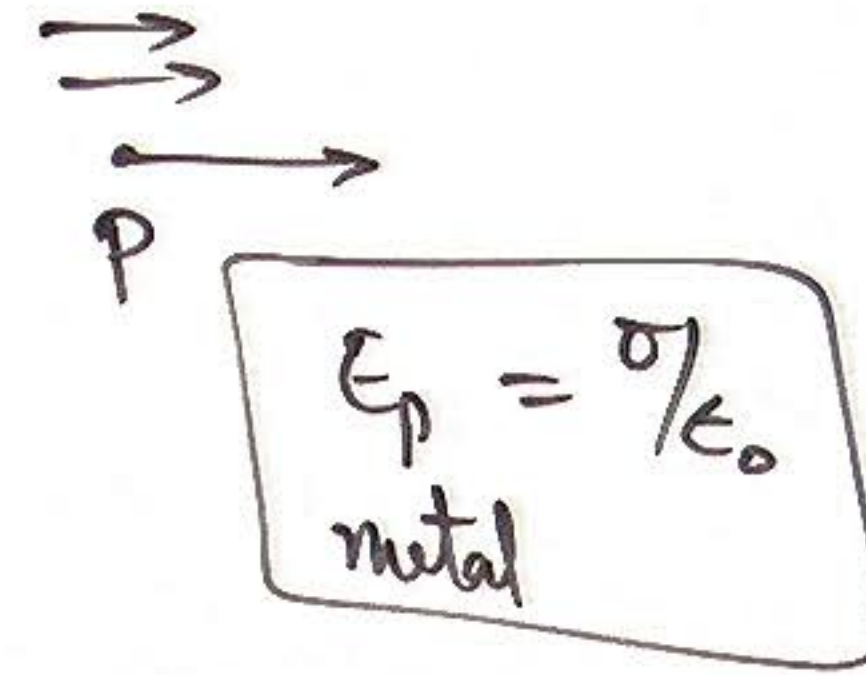
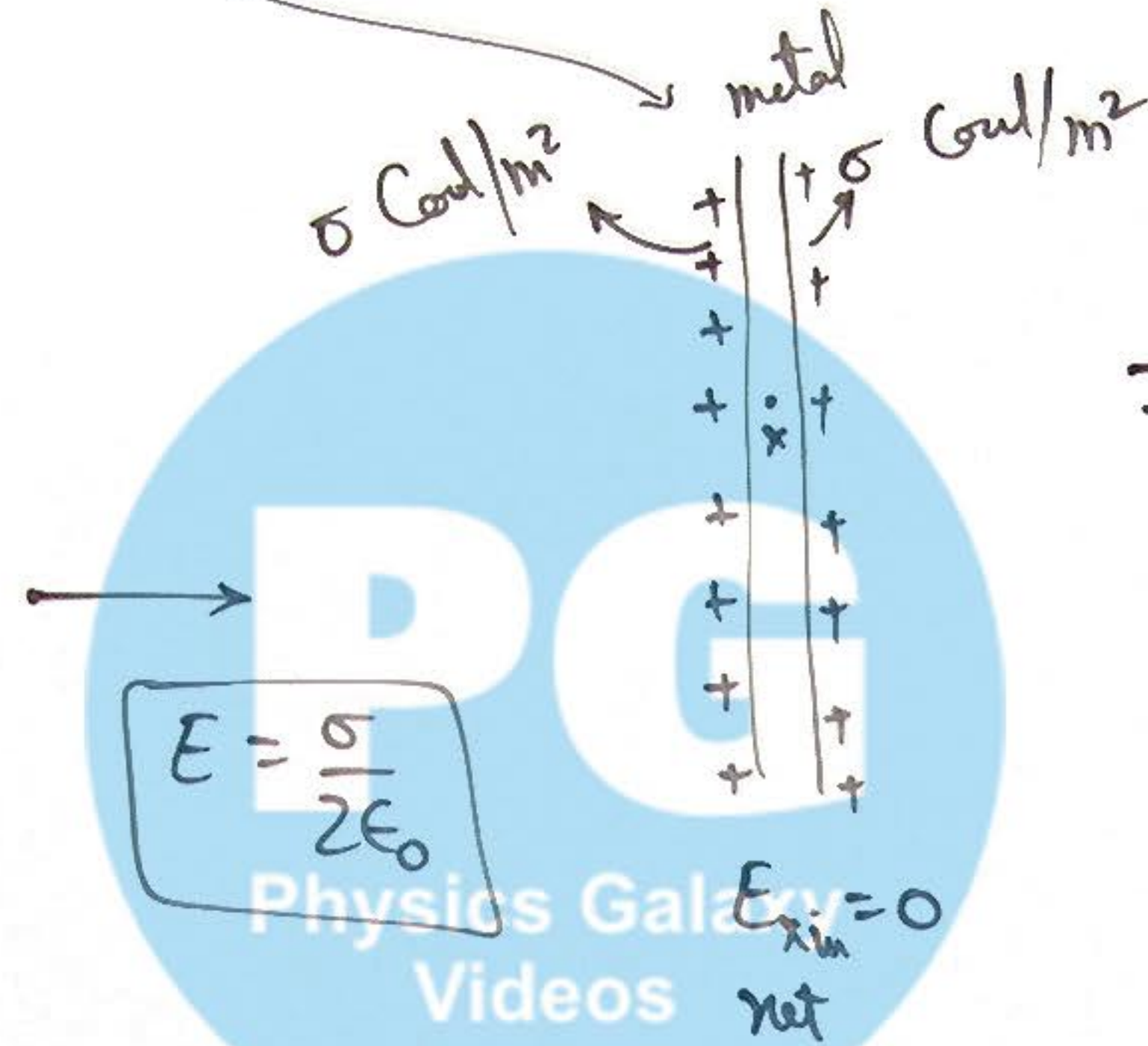
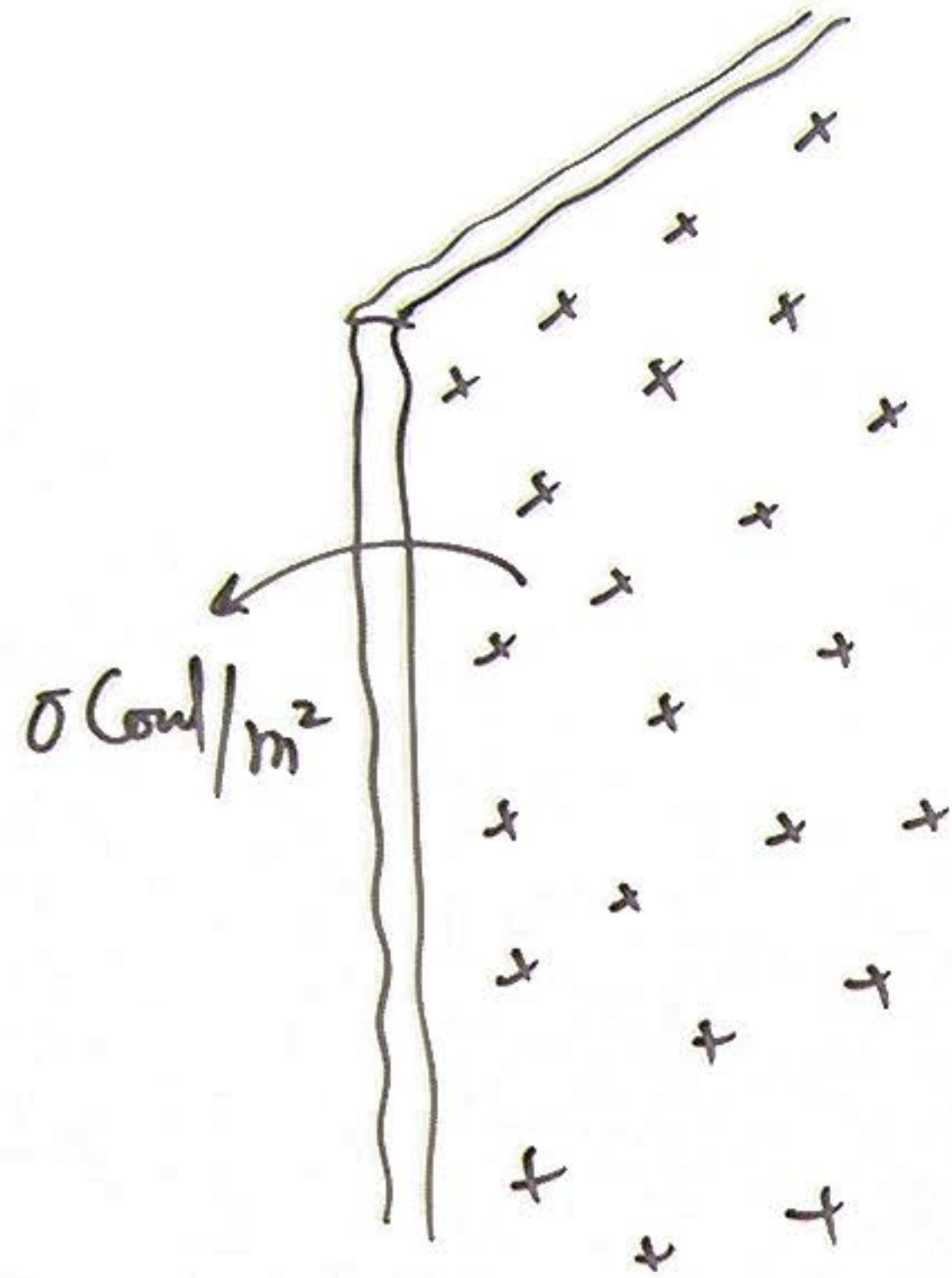
$$E_y = \frac{k\lambda}{r} [\sin\theta_1 + \sin\theta_2]$$



(16) EF due to a uniformly charged Disc



⑪ EF due to a charged Metal Sheet:

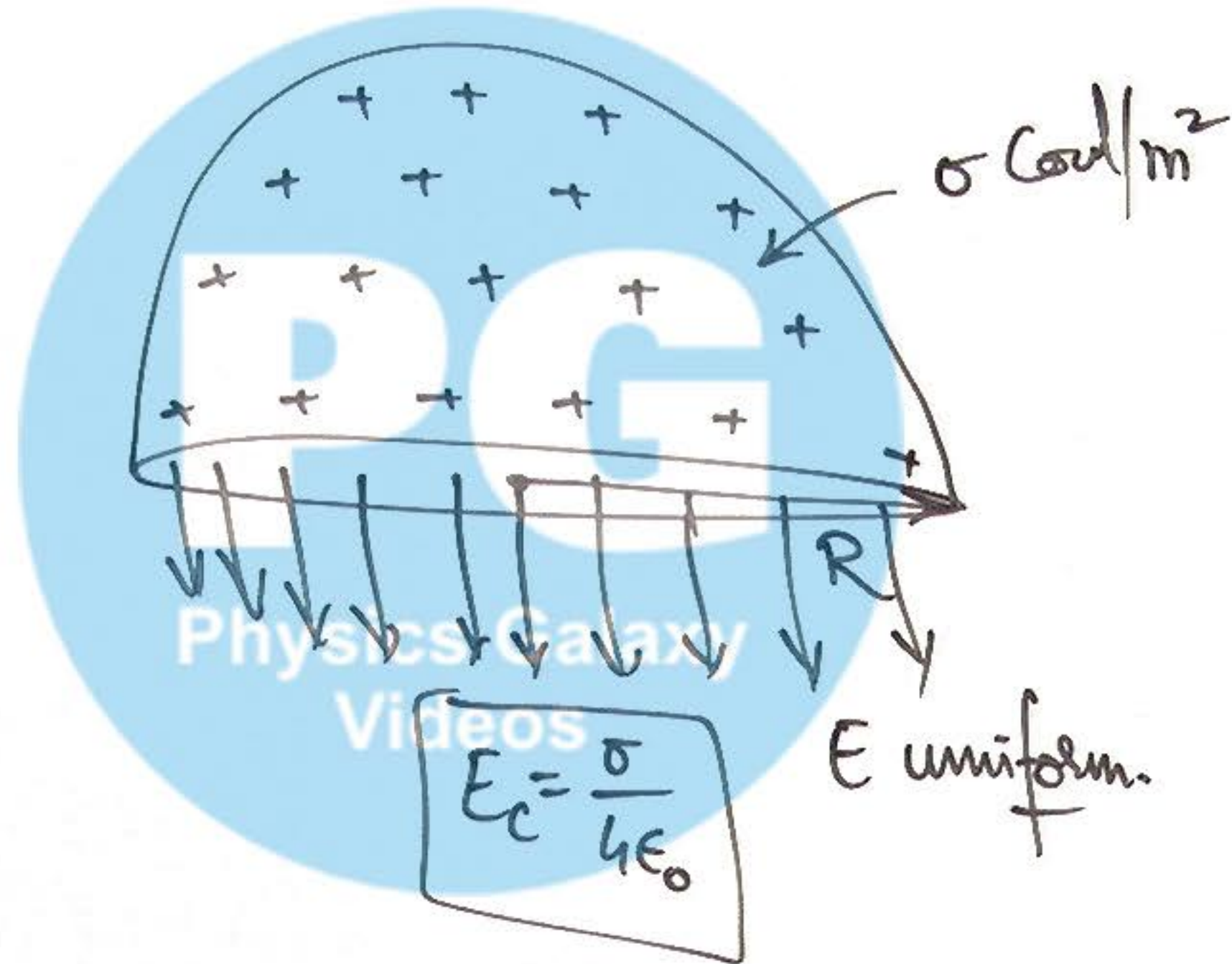


(18) EF due to a uniformly charged
hemispherical Cup :

flux through base

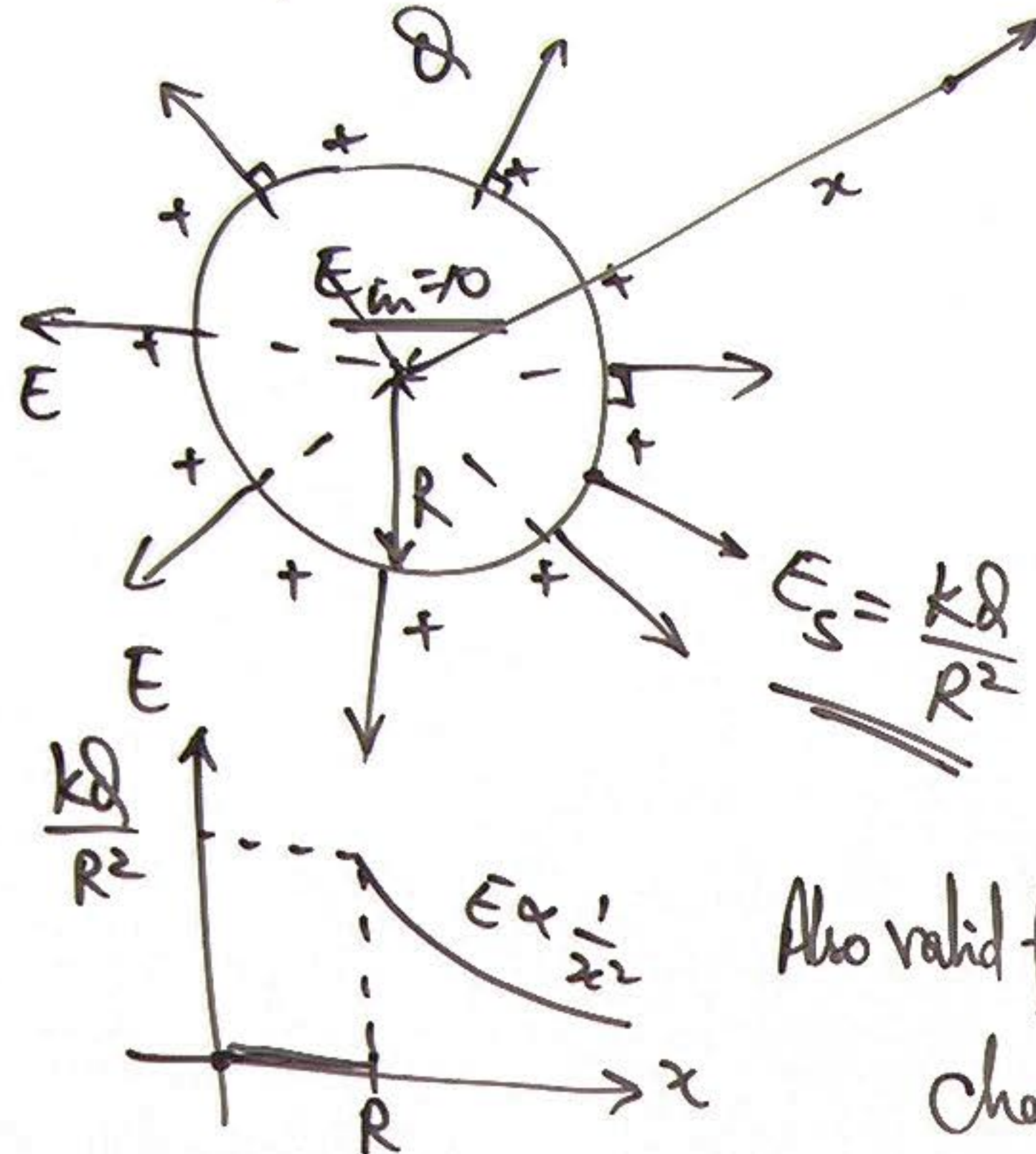
$$\phi = \vec{E} \cdot \vec{S}$$

$$= \frac{\sigma}{4\epsilon_0} \cdot \pi R^2$$



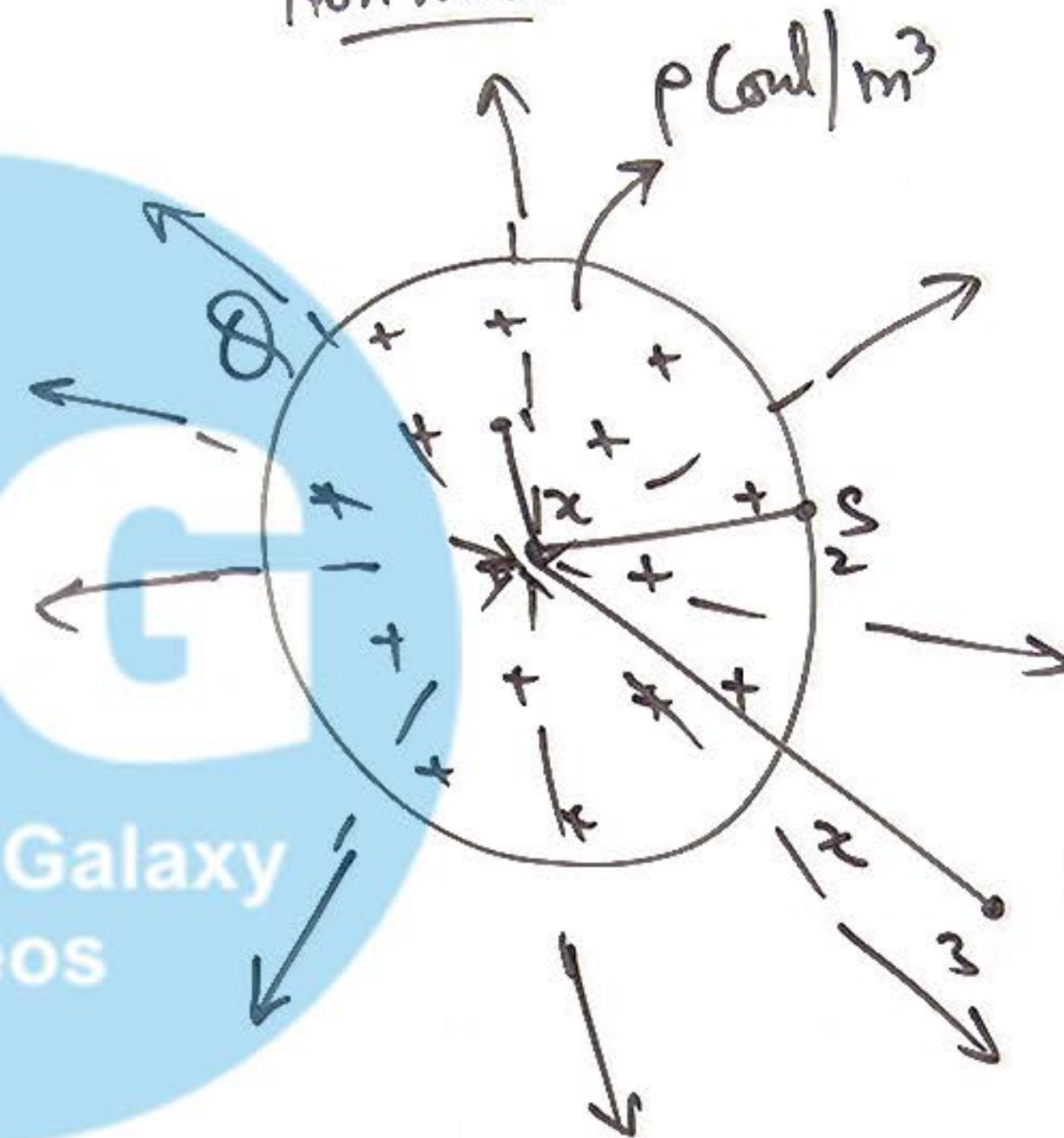
①9 EF due to a uniformly charged sphere:

Case-1
metal sphere



Also valid for uniformly charged hollow sphere.

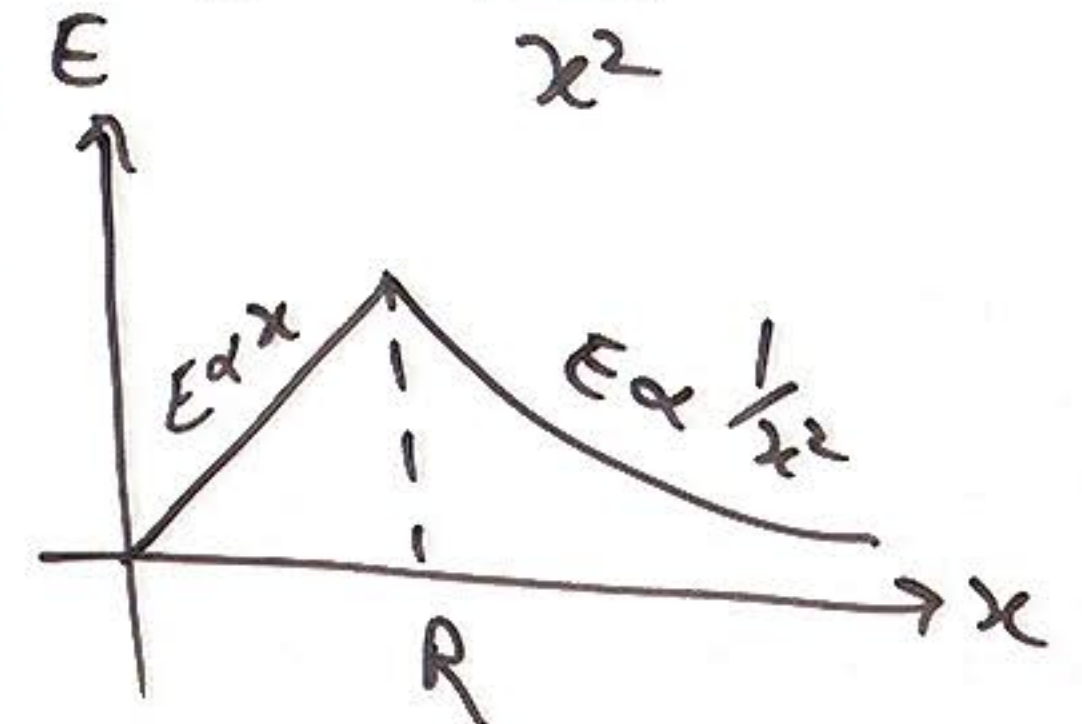
Case-2
Non metal



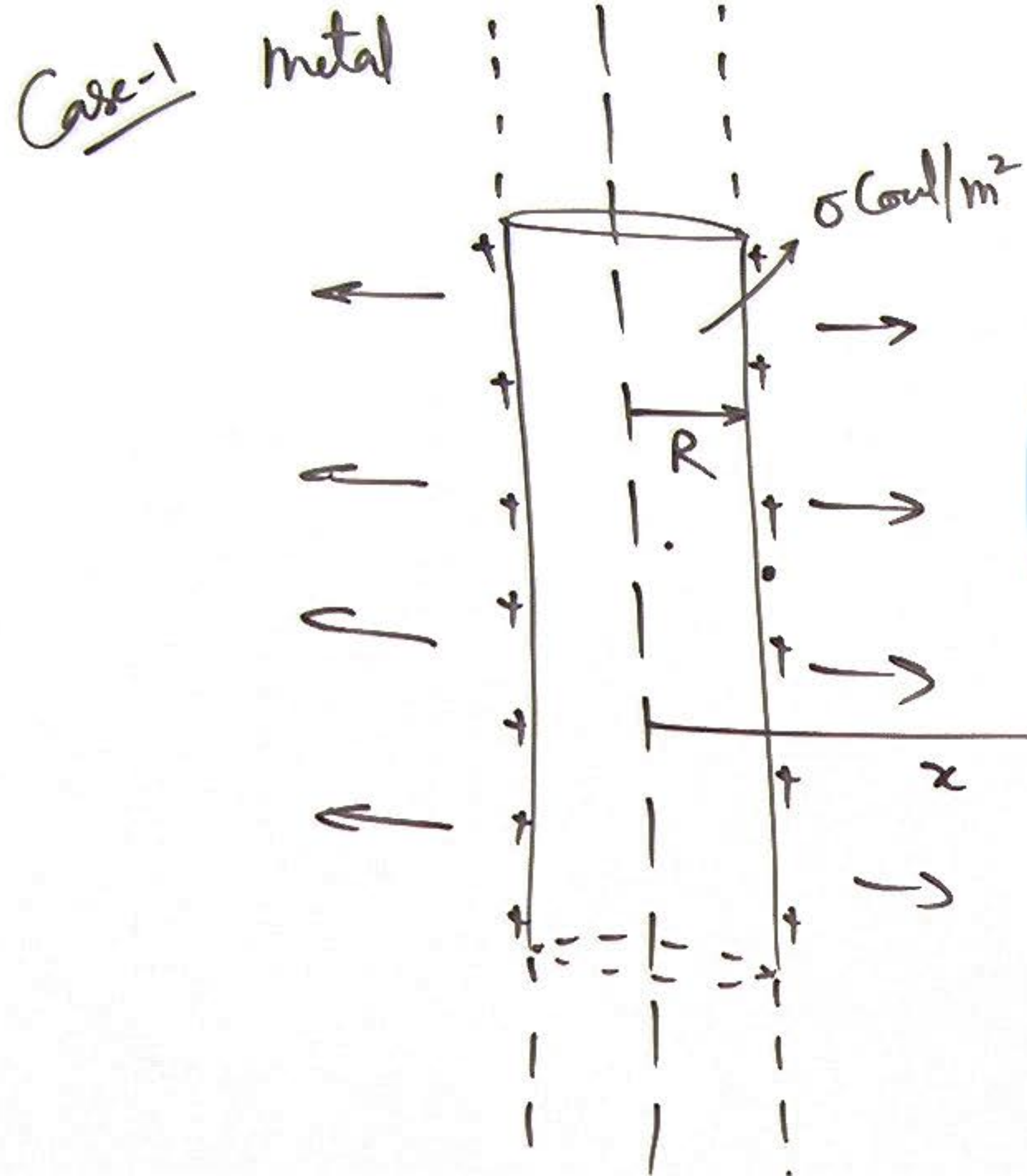
$$E_1 = \frac{kQx}{R^3} = \frac{\rho x}{3\epsilon_0}$$

$$E_2 = \frac{kQ}{R^2}$$

$$E_3 = \frac{kQ}{x^2}$$

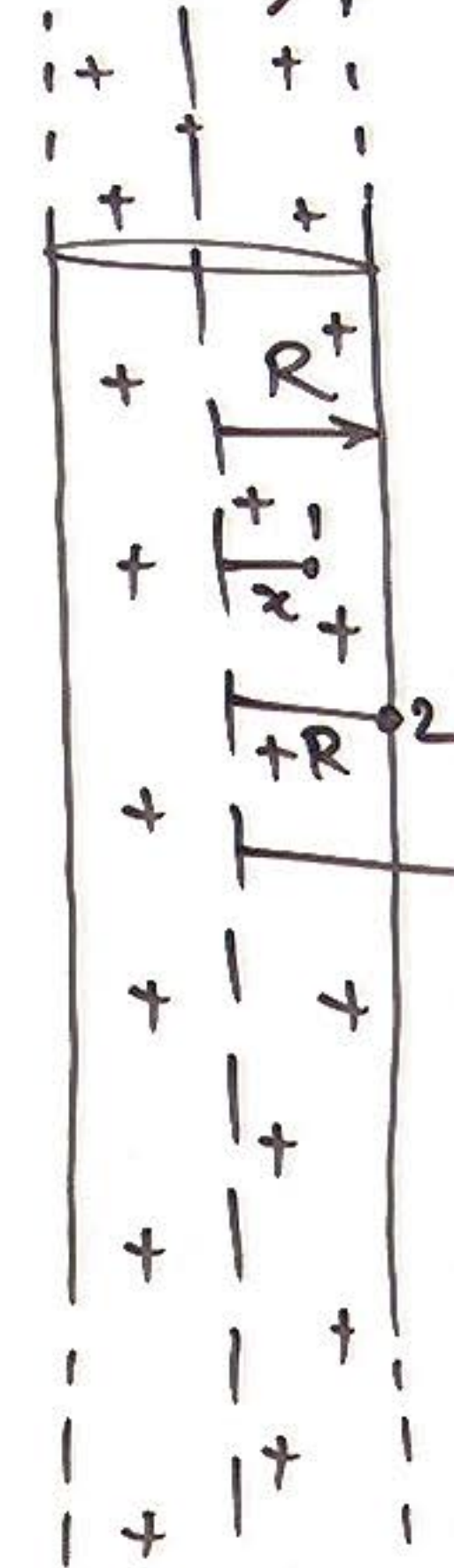


② EF due to a long uniformly charged cylinder :



$E_{in} = 0$
 $E_s = \frac{\sigma}{\epsilon_0}$
 $E_{out} = \frac{\sigma R}{\epsilon_0 x}$

Case-2 Non-metal $\rightarrow \rho \text{ Coul/m}^3$

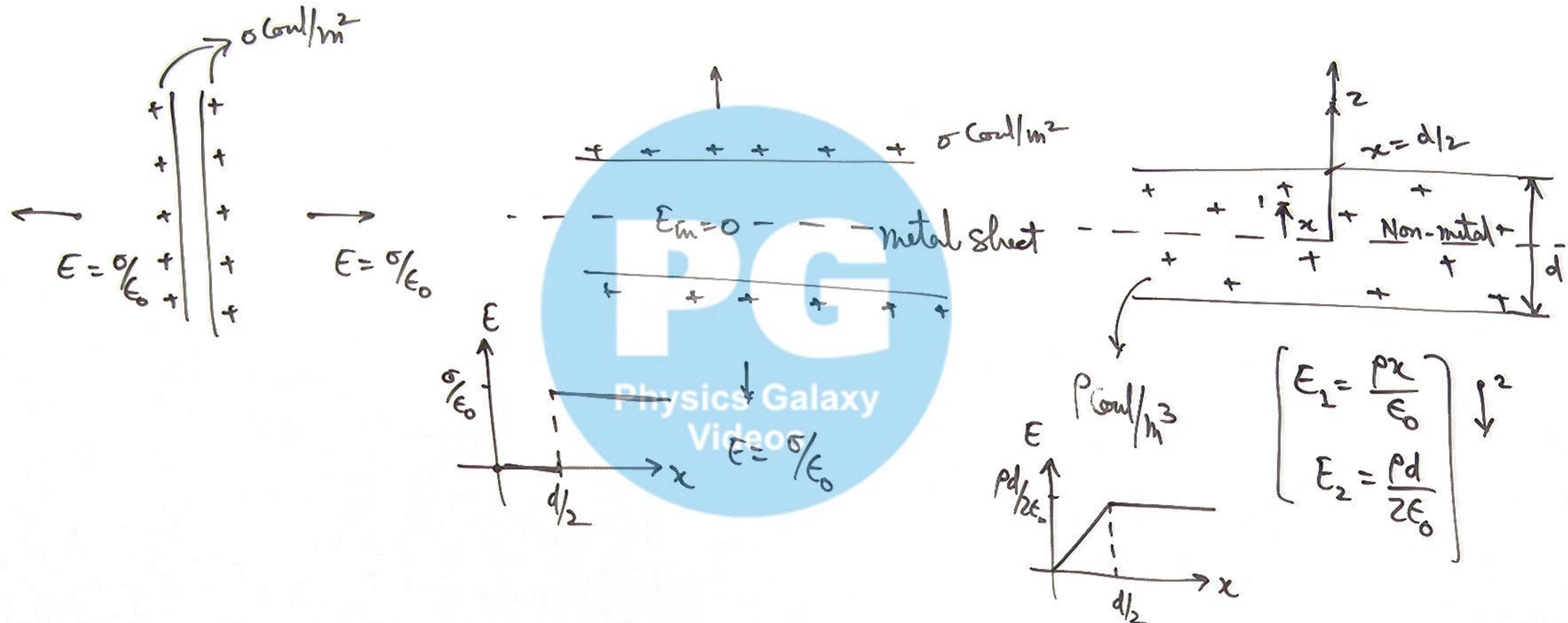


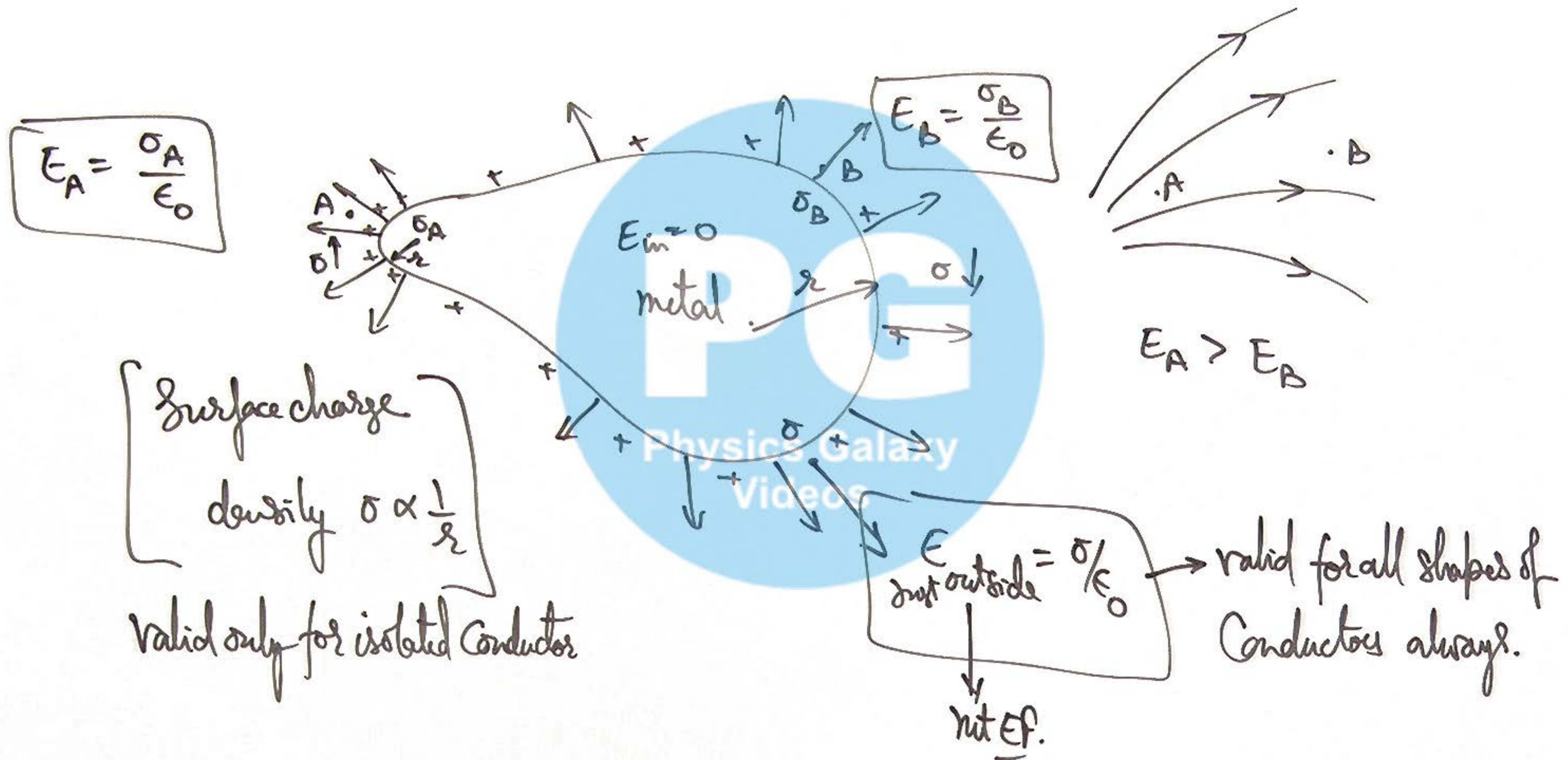
$$E_1 = \frac{\rho x}{2\epsilon_0}$$

$$E_2 = \frac{\rho R}{2\epsilon_0}$$

$$E_3 = \frac{\rho R^2}{2\epsilon_0 x}$$

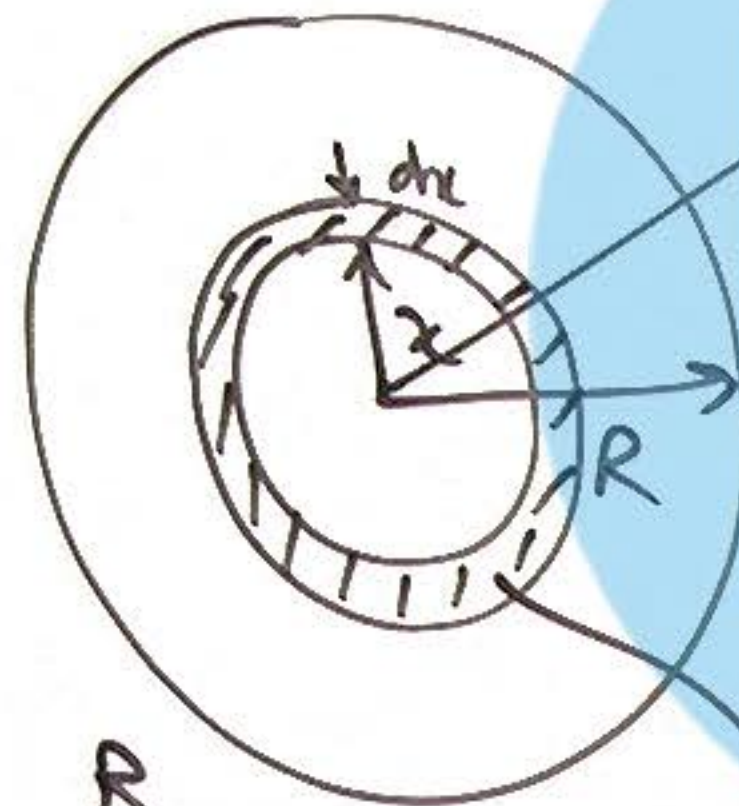
(21) EF due to a Thick charged Sheet:



(22) EF near a charged Conductor :

(23) EF due to Non-UniformCharge Distribution:We use integration

$$\rho(x) = \left(\frac{C}{x^2}\right) \text{Coul/m}^3$$

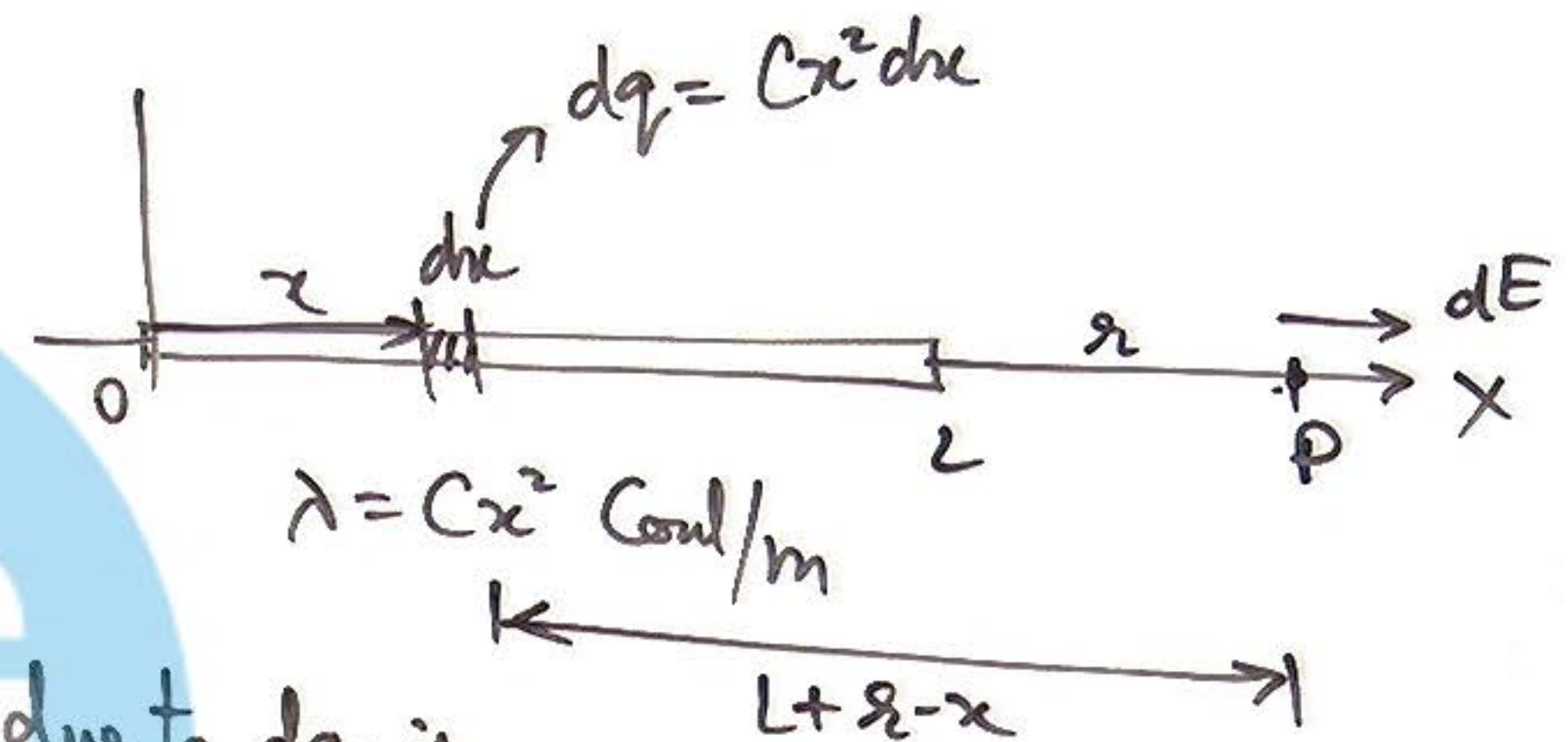
 $f(x)$ 

$$E_p = \int dE = \int_0^R \frac{k dq}{r^2}$$

line charge

 dE Physics Galaxy
Videos

$$dq = \rho(x) \times 4\pi x^2 dx$$

 dE due to dq is

$$E_{\text{net}} = \int dE = \int_0^L \frac{k dq}{(L+x-x)^2}$$